Examining LLM Assisted Rule Creation in WhatsApp Groups

Anonymous submission

Abstract

WhatsApp is the most popular global messaging platform, yet end-to-end encryption limits platform-level governance and shifts the responsibility to volunteer group admins, who often lack the resources and motivation to moderate harmful content. Although many communities rely on proactive governance through group rules, our analysis of 73,636 WhatsApp groups shows that only 0.03% articulate explicit rules, revealing a critical governance gap. Therefore, we investigate whether large language models (LLMs) can assist group admins in creating appropriate rules for their groups. Across two studies with 255 participants in India, we examine: (1) how varying levels of group information (group category only, message metadata only, all available data) influence admin's preferences for LLM-generated rules, and, (2) whether AI assistance produces higher-quality rules than human effort alone. In Study 1, admins preferred rules generated with full group context, while generic rules emerged as a viable alternative. In Study 2, crowd evaluations rated AI-assisted rules higher across five dimensions of perceived rule quality, and these rules covered nearly twice as many governance issues as human-written rules. Together, our findings provide the first empirical evidence that AI-assisted rule generation is feasible for encrypted messaging platforms.

1 Introduction

WhatsApp has established itself as the world's primary digital infrastructure for communication, connecting billions of users through end-to-end encrypted networks. However, this architectural focus on privacy creates a significant governance challenge: the platform has limited oversight into the problematic content, ranging from misinformation to harassment, that circulates within its groups (Banaji et al. 2019; Cartner-Morley 2022). Consequently, the burden of moderation falls entirely on group admins. These admins are typically volunteer users embedded in existing social hierarchies who often hesitate to penalize offenders or remove content due to the risk of damaging personal relationships (Shahid, Agarwal, and Vashistha 2025). Instead of punitive moderation, these volunteer moderators often rely on "softer, social approaches" such as articulating group rules (Kiesler et al. 2012) and educating members about these rules (Wohn 2019), strategies which have been shown to promote civil behavior (Dwork et al. 2024; Matias 2019).

However, despite the theoretical effectiveness of these soft governance strategies, naive approaches to implementing them have largely failed. Most platforms, including WhatsApp, barely provide any support for creating or enforcing such rules (Wohn 2019; Shahid, Agarwal, and Vashistha 2025). Moreover, existing research emphasizes improving rule enforcement, with little attention paid to supporting rule creation within online communities (Song et al. 2023; Wang et al. 2024). This gap is particularly glaring on end-to-end encrypted platforms, where the status quo is one of inaction. Our analysis of over 73,000 WhatsApp groups reveals that 99.72% of these groups lack descriptive rules, and only a fraction (0.03%) contain explicit rules. Prior research further shows that even when WhatsApp admins want to govern their groups, they lack the necessary support to create rules (Shahid, Agarwal, and Vashistha 2025). As a result, admins in formal work or study related WhatsApp groups appreciate the potential of AI assistance in crafting rules around issues that may otherwise get overlooked (Nayak et al. 2026).

Therefore, we explore if large language models (LLMs) can support WhatsApp group admins by proactively suggesting appropriate rules. While the generative capabilities of LLMs are well-documented, their application in encrypted environments remains unexplored, particularly regarding the trade-off between personalization and privacy. In particular, we ask the following questions:

- **RQ1**: What types of LLM-generated rules do WhatsApp group admins prefer? How does preference vary with the level of group context available to the LLM?
- **RQ2**: How do rules co-created with LLM assistance compare to the rules written by humans alone in terms of perceived quality and appropriateness?

We answer these questions through two empirical studies: first, by generating rules using varying levels of group context (generic, metadata-only, and full transcript) to understand admin preferences; and second, by comparing rules written by humans alone against those co-created with AI assistance to assess quality.

We conducted our studies in India, home to WhatsApp's largest user base, with 749 million active users (Gegenhuber et al. 2026). Prior work shows that social hierarchy in India often deters admins from moderating harmful content in close-knit groups (Shahid, Agarwal, and Vashistha 2025). The governance failures have often led to real-life harms and in some cases, group admins have faced legal consequences under the Indian Penal Code (Centre for Internet and Society 2017; Banaji et al. 2019). These dynamics make India a salient context for examining how AI assistance can support WhatsApp governance.

Our results demonstrate that AI assistance significantly improves the perceived quality of admin-articulated group rules in encrypted spaces. In our first study, admins significantly preferred AI-generated rules that used full group context, including anonymized conversation data, over rules generated using only conversation metadata. We did not find any significant preference for contextual rules over generic rules, which were produced using group type as the only context. However, our qualitative analysis suggests that contextual information was especially useful for niche groups, such as study or work related groups. In Study 2, rules co-created with LLM assistance were significantly preferred over rules written by humans alone across multiple evaluation dimensions.

This work provides the first empirical evidence that AI-assisted rule articulation is feasible in encrypted messaging platforms and produces rules that admins perceive as useful and of high quality. We demonstrate and argue that the bottleneck in creating safer online spaces is often not a lack of intent, but a lack of support, and that LLMs can effectively aid admins in creating rules for their groups. By identifying how different levels of data access influence the quality of governance tools, we offer a blueprint for messaging platforms to integrate safety features that respect the delicate balance between user privacy and community health.

2 Related Work

Rules and Norms Across Platforms. Rules in online communities serve as the primary mechanism for promoting responsible behavior and maintaining social order (Matias 2019). The forms and functions of these rules vary significantly across platforms. Early governance on Wikipedia emphasized “constitutional” principles, such as neutrality and consensus—alongside specific guidelines for editing and administrative authority (Bruckman 2022; Hwang and Shaw 2022). On Reddit, rules are predominantly *restrictive*, specifying prohibited behaviors, though some communities adopt *prescriptive* norms that encourage positive conduct (Fiesler et al. 2018; Leibmann et al. 2025). Chandrasekharan et al. (2018) further categorized these rules by their scope, distinguishing between platform-wide *macro* rules, commonly shared *meso* rules, and community-specific *micro* norms.

Despite the importance of these rules, most platforms offer little support for creating them (Wohn 2019). Admins often rely on trial and error, personal intuition, or copying legitimate-sounding rules from other groups (Chandrasekharan et al. 2019; Seering et al. 2019; Sultana et al. 2022). Consequently, many groups launch with generic rules focused vaguely on “civility” (Reddy and Chandrasekharan 2023), while more specific norms only emerge reactively after disruptive events, such as surges in misinformation (Chadwick, Hall, and Vaccari 2025) or the influx of low-quality AI-generated content (Lloyd et al. 2025).

This lack of scaffolding is particularly critical on end-to-end encrypted (E2EE) platforms like WhatsApp. Unlike public forums where moderation can be centralized or crowd-sourced (e.g., downvoting), E2EE architecture restricts platform-led oversight, shifting the entire burden of moderation onto group admins (Shahid, Agarwal, and Vashistha 2025). Recent surveys of E2EE governance highlight that this privacy-safety trade-off often leads to inaction; without tools to detect harmful content or define norms, admins frequently abdicate their moderation duties entirely to avoid social friction (Scheffler and Mayer 2024). Our work addresses this gap by proposing a proactive, privacy-preserving method to help admins define these norms before violations occur.

AI in Online Governance. Social computing research on AI governance has historically focused on *enforcement* rather than *creation*. Extensive work has been done on developing AI tools to flag rule-violating content (Kumar, Hashem, and Durumeric 2024), detect toxic threads before they escalate (Schluger et al. 2022; Choi et al. 2023; Liu, Choi, and Chandrasekharan 2025), and triage complex cases for human review (Koshy et al. 2025). While effective for public platforms, these enforcement-heavy approaches are technically infeasible on WhatsApp due to encryption and socially undesirable in small, relational groups.

A nascent body of work has begun to explore AI for rule *creation*. Xiang et al. (2023) explored how LLM-based agents can simulate community reactions to propose robust rules, while Feng et al. (2026) and Wang et al. (2024) developed interactive systems to help community members negotiate and craft moderation policies. However, these studies predominantly focus on public, transparent communities like Reddit or Discord. Our work extends this domain into the encrypted, high-context environment of WhatsApp, specifically investigating how the *amount* of private context shared with an AI impacts the perceived utility of its governance suggestions (Study 1).

Human-AI Collaborative Writing. Writing governance rules is fundamentally a creative and rhetorical task. Therefore, our work also builds upon the rich literature of Human-AI collaborative writing. Large Language Models have been shown to effectively serve as “ideation partners,” helping humans overcome writer’s block and explore diverse perspectives they might have missed (Buschek, Zürn, and Eiband 2021; Lee, Liang, and Yang 2022). Recent studies on co-writing suggest that while humans often retain control over the final intent, they rely on AI to handle the structural and stylistic execution of text (Gero, Liu, and Chilton 2022; Reza et al. 2025).

However, this collaboration is not without friction. Users must balance the ease of automation with the need to maintain their own voice and agency (Agarwal, Shahid, and Vashistha 2024). In the specific context of governance, this “voice” is crucial because rules must feel legitimate to the community. By comparing human-only rule drafting against an AI-assisted workflow (Study 2), we contribute to this literature by quantifying not just the *efficiency* of AI collaboration (e.g., ease of writing), but the *quality* of the rules (e.g., enforceability and tone) in a high-stakes social context.

Synthesizing these diverse strands of research, our work bridges the gap between algorithmic governance and AI-assisted writing within the unique constraints of end-to-end encrypted platforms. We advance this domain through two primary contributions: first, we empirically map the trade-off between privacy and utility by investigating how varying levels of group context, from generic labels to granular metadata, shape admin preferences for LLM-generated rules. Second, we move beyond mere feasibility to assess the *qualitative* impact of AI collaboration, rigorously compar-

Table 1: Content analysis of WhatsApp group descriptions. Categories are mutually exclusive.

Content Category	N	%
Generic content (greetings, emojis, short misc. text, etc.)	99	48.5
URL/link only (no explanatory text)	50	24.5
Behavioral rules/guidelines	20	9.8
Event/schedule information	13	6.4
Group purpose/mission	12	5.9
Contact information (phone/email/handles)	5	2.4
Promotional/commercial content	3	1.5
Registration/signup information	2	1.0

ing the efficacy, tone, and enforceability of rules co-created with LLMs against those drafted by humans alone.

3 The Status Quo of WhatsApp Group Rules

To motivate our investigation, we first examined to what extent WhatsApp groups set any rules. We analyzed group metadata collected via WhatsApp Explorer (Garimella and Chauchard 2025), a privacy-preserving web application that enables users to voluntarily donate messages and metadata of their chosen WhatsApp groups. The dataset provided by Garimella et al. contains *73,636* unique groups from 5,215 users across multiple regions, with group names in English (69%), Hindi (12%), Spanish (4%), and other languages, reflecting WhatsApp's global user base.

Rules within Group Descriptions. Unlike subreddits and Facebook Groups, which offer a dedicated space to post community rules (Fiesler et al. 2018; Meta 2018), WhatsApp does not offer such an option. While WhatsApp recommends using 'Community Descriptions' for explaining the purpose of a WhatsApp Community and 'Announcement Group' to communicate the rules (WhatsApp LLC 2022), there is no such public facing suggestions for WhatsApp groups. WhatsApp currently offers two key ways to share information with the group for long-term: pinned messages, which let a member pin an important message for quick access later, and the group description, which provides a 512-character limit to share basic group information.

The WhatsApp Explorer dataset did not include pinned messages. So, we only analyzed the available group description. We found that 99.72% of the groups (73,432 of 73,636) have no description at all. Among the 204 groups with descriptions (0.28%), content varies widely (Table 1): over a third contain miscellaneous text or URLs without substantial context, while only 9.8% (20 groups—0.03% of all groups) contain any explicit rule at all.

Characterizing Existing Rules. We manually identified 16 unique group descriptions containing rules across 20 groups. Using prior taxonomy on Reddit rules (Fiesler et al. 2018; Leibmann et al. 2025) that characterizes rules based on tone, target, topic—properties that are platform-agnostic– we analyzed each group description (Table 4 in Appendix).

The majority of rules were framed as *restrictive* (75%), specifying prohibited behaviors (e.g., "NO MEMES. NO SOLICITATIONS"), consistent with patterns observed on Reddit where restrictive rules also predominate (Leibmann et al. 2025). Rules primarily target *post content* (11), restricting off-topic discussion and spamming. We identified rules beyond existing taxonomies, including references to external norms like Chatham House Rules, Google Doc links containing organizational norms, and WhatsApp-specific privacy rules prohibiting forwarding or taking screenshots.

These observations indicate that the majority of WhatsApp group admins almost never create rules, motivating us to investigate whether LLMs can support admins in proactively creating appropriate rules for their groups. For this, we conducted two studies. In Study 1, we investigate whether WhatsApp admins find LLM-generated rules useful at all, by examining how varying levels of group context (generic, metadata-only, full context) affect rule preferences (RQ1). In Study 2, we compare rules written by humans with and without AI assistance to assess the potential of human-AI collaboration in rule creation (RQ2). Both studies received IRB exemption at [University Name].

4 Study 1: Preference for LLM-Generated Rules

Study 1 investigated how different levels of group context influence the quality and selection of LLM-generated rules for WhatsApp groups. We employed a mixed experimental design, with a within-subjects factor where for each participant the rules were generated by LLM with varying levels of group context (generic, metadata-only, or full context), and a between subjects factor, where the LLM-generated rules were either presented as is (Study 1A) or with rationale (Study 1B). This design allowed us to examine how group context shapes LLM-generated rules and whether exposing participants to AI's rationale influences their selection of rules.

We tested three distinct group contexts considering the practical constraints in real-world governance scenarios. While full message provides rich context for rule generation, many groups may be hesitant to share complete conversation histories with AI systems due to privacy concerns. The metadata-only condition represents a privacy-preserving alternative that only leverages patterns in group activity without exposing actual message or user information. The generic condition serves as a baseline representing rules generated purely from the model's knowledge base about such group types. By comparing rule selection across these conditions, we can assess which approach yields better group rules that participants find useful.

Additionally, we controlled the presentation of LLM-generated rules (Study 1A vs. 1B) to assess whether the presence of explanations affect what rules participants select for their groups.

4.1 Procedures for Study 1

We recruited participants via Prolific from India. A summary of our procedure is described below:

1. Participants uploaded a transcript of group chat for one of their active WhatsApp groups (at least 50 messages in the last 30 days). We provided detailed instructions on how to download group chat for different devices.
2. Participants were presented with the rules generated independently by an LLM with differing access to the group's context.
3. Participants were asked to select up to 5 rules that they found appropriate for their group (minimum of 1).
4. We showed participants a random rule that they selected for their groups and asked to provide reason. We repeated the same process for one random non-selected rule.

We provide compensation information and other instructions in Figure 19. Since WhatsApp generates chat transcripts in a fixed format for different platforms (android, ios/MacOS, windows), we removed all personal data using this pattern (e.g., group members' names, contact numbers, media files etc.) before prompting the LLM with the chat transcript. We did not store the anonymized transcript in our systems to protect participant privacy. The complete procedure for Study 1 is depicted in Figure 2.

Group Contexts For each uploaded transcript, we generated three different types of rules, where the LLM had varying access to the group's context:

In the **generic condition**, the LLM received only the group category that the participant selected from a predefined list (Family, Friends, Work, Community, Hobby/Interest, Support, Educational, or Other with free-text input). The prompt explicitly instructed the model to generate rules solely based on its prior knowledge of what constitutes appropriate norms for that type of group. This condition represents the baseline of LLM-generated rules, relying entirely on the model's training data and the group type.

In the **metadata-only condition**, the LLM received aggregated statistics and per-message metadata but no actual message content. The exact metadata information passed to the LLM is available in appendix B.2. The prompt explicitly instructed the model to consider behavioral patterns present in the metadata when suggesting rules. This condition tests whether behavioral cues from the metadata—such as message frequency, the time of messaging, number of external links or media, etc. can lead to useful rules without compromising the group's privacy.

In the **full context condition**, the LLM received all metadata along with the anonymized chat transcript. Each message was represented as a JSON object containing timestamp, sender pseudonym, media type, and the full message with redacted PII. The prompt instructed the LLM to analyze the transcript for recurring topics, conflicts, emergent norms, and problematic behaviors, and to generate rules strictly grounded in these observations. Complete instructions for this condition are also provided in appendix B.2.

Rule Presentation For each condition, we separately instructed the LLM to generate 5 distinct rules along with rationale targeting different aspects of the group. The rationale was hidden from participants in Study 1A and shown in 1B. To ensure that similar rules from different conditions were not presented separately, we processed all the 15 (5×3 conditions) rules and corresponding rationale and merged based on contextual similarity of the rules in Study 1A and the contextual similarity of both the rules and rationale in 1B. Two authors independently verified that the merging occurred as expected in a pilot study of 16 participants. If a participant selected a merged rule, it was counted against all the categories it was merged from. Critically, merging was permitted only across conditions (e.g., a generic rule with a contextual rule, never two generic rules together), so that conceptually overlapping rules independently generated by different conditions would not inflate any single condition's representation in the choice set. A merged rule thus carries tags from all contributing conditions, and a participant's selection of a merged rule counts toward each contributing condition.

We capped the number of rules presented to the participants at 12 (i.e., after merging if we still had 15 rules, we would drop 3; if we had 14, we would drop 2). We implemented a look-ahead dropping logic to make sure that after dropping the total number of merged rules across conditions were the same. More details on the implementation of merging and dropping along with worked examples are present in appendices B.2 and B.4 respectively. The distribution of total rules presented to participants after merging and dropping are presented in Figures 6 and 11 respectively, in the Appendix. We used Gemini 2.5 Pro to generate the rules because it allows large input context length and Gemini 2.5 flash for merging rules to get faster responses.

Privacy and Ethics. Our IRB-approved protocol follows the established data-donation methodology for end-to-end encrypted platforms (Ohme and Araujo 2022; Garimella and Chauchard 2025), where universal third-party consent is structurally infeasible. For study 1, we incorporated client-side PII removal and sender pseudonymization using the predictable structure of WhatsApp chat exports. We redacted URLs, emails, and phone-like sequences using regex, and media attachments are discarded entirely. We also present the anonymized chat to the participant to review incidental personal content that automated redaction may miss. We only stored the generated rules and discarded the chat file to minimize the amount of private data stored. Our anonymization is best-effort and operates under the public-interest research framework adopted by the WhatsApp data-donation literature. For more information on threat models and leakage, consider Appendix B.3.

Data Collection and Analysis We hypothesized whether the proportion of rules selected from full context condition would be higher than both metadata-only and generic rules condition, across both 1A and 1B. We recruited 62 participants for Study 1A and 55 for 1B after we conducted a power analysis for our hypothesis with a pilot of 16 participants. More information about the power analysis is provided in Appendix B.1.

We employed paired statistical tests to compare the percentage of selected contextual rules by each participant against that of generic and metadata-only conditions. We applied Bonferroni correction for two planned comparisons per Study ($\alpha_{adjusted} = .025$). To control for potential confounding by group category (e.g., Personal, Professional, Learning), we tested whether the preference for different categories of rules varied across group types using one-way ANOVA on difference scores. Both parametric (paired t-tests) and non-parametric (Wilcoxon signed-rank tests) approaches yielded consistent results; we report the parametric analyses with non-parametric tests as robustness checks.

4.2 Study 1: Findings

LLM-Generated Rules We manually reviewed the LLM-generated rules in Study 1 and identified 32 unique rules. We also analyzed the tone, target, and topics of these rules following the taxonomy developed by Leibmann et al. (2025). We noticed that LLM framed 90% of the rules as prescriptive, in contrast to the predominantly restrictive rules usually crafted by WhatsApp admins (Section 3) or Reddit moderators (Fiesler et al. 2018). These LLM-generated rules targeted post content (37.5%), post formatting (25%), or user activities (37.5%). Majority of these rules focused on peer-engagement (46.9%), and discouraged sharing of spam and low-quality content (28%).

Additionally, we reviewed the LLM-generated rationale for these rules to determine the factors the model considered when suggesting the rules. For rules in generic condition, the LLM prioritized prosocial goals, such as preserving group trust, harmony, privacy and security, mutual respect, and integrity (e.g., "Do not share content from this group outside without consent"). For metadata-only condition, LLM considered the number of messages sent by group members, number of deleted messages, number of media content or external links, and message length, among others. For example, LLM suggested the rule "Review before posting messages to avoid deleting later"–noticing several deleted messages in the group. In full context condition, LLM integrated the actual message content. For instance, LLM recommended "Share information clearly to coordinate large group events"–after noticing back and forth and confusion from last minute changes in the event planning.

Rule Preferences In Study 1A (**No Rationale Condition**; $N = 62$), participants selected contextual rules at a higher rate ($M = .51$, $SD = .20$) than both generic ($M = .41$, $SD = .22$) and metadata-only rules ($M = .33$, $SD = .19$). The preference for contextual over generic rules was directionally consistent but did not survive Bonferroni correction, $\Delta M = .10$, 97.5% CI $[-.001, .19]$, $t(61) = 2.21$, $p = 0.030$, $d = 0.28$, 95% CI $[0.03, 0.54]$. The preference for contextual over metadata-only rules was statistically significant, $\Delta M = .18$, 97.5% CI $[.09, .27]$, $t(61) = 4.37$, $p < .001$, $d = 0.56$, 95% CI $[0.29, 0.82]$. Paired permutation tests on the mean differences—which we adopt as a non-parametric robustness check because the discrete bounded nature of the selection proportions (i.e, one of 0, 0.2, 0.4, 0.6, 0.8, 1.0) produces a large fraction of tied differences (0)—yielded consistent conclusions (contextual vs. generic: $p = .046$; contextual vs. metadata: $p < .001$).

The effect did not significantly vary across group types for either comparison (contextual vs. generic: $F(4, 57) = 0.97$, $p = .434$; contextual vs. metadata: $F(4, 57) = 0.84$, $p = .506$), indicating that participants' preferences across conditions were consistent regardless of group type.

Similarly, in Study 1B (**Rationale Condition**; $N = 55$), participants selected contextual rules at the highest rate ($M = .50$, $SD = .22$), followed by generic ($M = .46$, $SD = .17$) and metadata-only rules ($M = .39$, $SD = .19$). The preference for contextual over generic rules was not statistically significant, $\Delta M = .04$, 97.5% CI $[-.06, .14]$, $t(54) = 0.96$, $p = 0.341$, $d = 0.14$, 95% CI $[-0.14, 0.41]$. However, the preference for contextual over metadata-only rules was significant after Bonferroni correction, $\Delta M = .12$, 97.5% CI $[.01, .23]$, $t(54) = 2.48$, $p = 0.016$, $d = 0.35$, 95% CI $[0.07, 0.63]$. Permutation tests yielded consistent conclusions (contextual vs. generic: $p = .40$; contextual vs. metadata: $p = .022$). As in Study 1A, the effects did not significantly vary by group category (contextual vs. generic: $F(4, 50) = 1.36$, $p = .264$; contextual vs. metadata: $F(4, 50) = 0.93$, $p = .453$).

We confirmed these findings under a Monte Carlo sensitivity analysis of the rule-merging and dropping procedure, combining sampling and counterfactual uncertainty via Rubin's Rules; the contextual-versus-metadata results remained significant under the pooled inference, with study 1B's comparison borderline at the Bonferroni-adjusted threshold (see Appendix B.5).

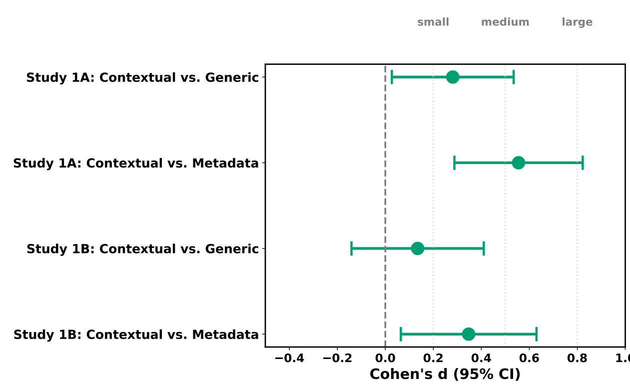

Figure 1: 95% Confidence Intervals (CIs) for effect sizes for hypotheses in Study 1.

What Makes Rules Desirable? To complement our hypothesis, we conducted an exploratory semantic analysis of the generated rules to understand *which types of rules* participants preferred and how rule content related to selection behavior. This analysis provides interpretive context for the quantitative findings and surfaces patterns that may inform future rule generation systems.

We embedded all rules (701 in Study 1A; 539 in Study 1B) using the GTE-Qwen2-7B-instruct model (Li et al. 2023), a state-of-the-art sentence embedding model optimized for semantic similarity tasks[1] We ended up with 66 clusters for Study 1A (Figure 5) and 48 clusters for Study 1B (Figure 10). We quantified cluster quality using *coherence*, defined as the mean pairwise cosine similarity among all intra-cluster rules. Higher coherence indicates that rules in a cluster express semantically similar ideas. We also provide the themes associated with each cluster and the composition of rules in Tables 7 and 12.

Rules with High Selection Rates Several rule themes were consistently selected highly across both studies 1A and 1B (Figures 7, 12).

Coordination and Relevance: Rules offering concrete coordination procedures were among the most selected: finalizing group decisions through polls (Study 1B, C22: 82%), confirming event attendance (Study 1A, C14: 77%), and settling shared expenses by deadlines (Study 1A, C50: 62%). Similarly, rules protecting group focus: keeping discussions "strictly related to the group's professional purpose" were selected highly (Study 1A, C48: 71%; Study 1B, C35: 57%). These rules were generated both as part of generic and contextual conditions, suggesting their broader applicability to group governance.

Information Quality: Rules against spam and misinformation were consistently favored: avoiding "unverified news, chain messages, or unsolicited advertisements" (Study 1A, C9: 67%; Study 1B, C32: 44%) and verifying information before sharing (Study 1A, C62: 75%). These often originated as generic rules, indicating that some governance concerns transcend group-specific contexts.

Group-Specific Procedures: The highest selection rates were associated with contextual rules addressing specific needs that can be only inferred from actual group chat. For example, using the group's designated common language (Study 1A, C40: 100%), adhering to assignment deadlines in a school-based group (Study 1A, C2: 100%), and sharing relevant job opportunities (Study 1A, C53: 78%). These indicate that such niche rules can only be suggested when the full group context is accessible to the model.

Rules with Low Selection Rates Certain types of rules were consistently avoided:

Timing and Formatting Restrictions: Rules about late-night messaging were among the least selected (Study 1A, C7/C39: 10–15%; Study 1B, C7: 38%), despite high coherence (0.94–0.95). Similarly, readability suggestions ("consider using paragraphs") achieved 0–20% selection (Study 1B, C23; Study 1A, C27). Participants may view these rules as pedantic, difficult to enforce, or redundant given technical affordances like muting notifications.

Participation and Consolidation: Rules encouraging "all members to contribute" showed consistently low selection (Study 1A, C55: 31%; Study 1B, C19: 27%), as did message consolidation rules (Study 1A, C17: 28%; Study 1B, C10/C47: 0–27%). Both were predominantly generated from metadata (88–100%). While the LLM correctly identified participation imbalances and fragmented messages from metadata, participants did not want to codify these observations into explicit norms.

Reason for Rule Selection We analyzed the open-ended responses participants provided for both selecting and not selecting certain rules. Participants selected rules that restricted spam, low quality, and divisive content, focused on improving group functionality, peer engagement, drawing boundaries, and respecting others. One participant who selected the rule discouraging personal attacks commented:

> *"This group is a fun and safe space for like minded gamers who like playing Fortnite. Jokes and comments should be limited to the game ane not to anyone's religion. None's sentiments should be hurt."*

Another participant selecting a rule about keeping discussions professional noted:

> *"It matters because the WhatsApp group is for professional work and too many casual chat or memes hide important job-related updates. When the group stays focused, everyone can quickly see tasks, deadlines and other details."*

In contrast, for non-selected rules participants reported that the rules could restrict freedom of speech, conflict with existing norms and group dynamics, or addressed issues they did not see as relevant. One participant explained that they did not choose the rule asking members to send a single message instead of multiple short replies, because:

> *"It's a family group, and it's informal and fun. I don't want to enforce a rule such as this which will prevent people from expressing themselves, especially senior citizens who might face an issue to follow these instructions."*

Participants also rejected rules they viewed as unnecessary or already implicitly followed:

> *"It is already understood. All the members are smart enough to follow this."*

Notably, rules encouraging participation were often rejected as potentially counterproductive:

> *"It should not be a compulsory rule because it's everyone's personal choice if they want to speak or not. Rules must be designed in a way that they should not be strictly imposing one's choice of talking. If it's a friendly group, people will automatically talk—don't want to impose these kinds of rules."*

Overall, these findings indicate that taking group context into account can lead to rules that the community find useful for their WhatsApp groups.

5 Study 2: Rules With or Without AI Assistance

While Study 1 examined LLM-generated rules for participants' own WhatsApp groups based on uploaded transcripts,

[1] We compared three methods to find thematic clusters: agglomerative clustering (average linkage), Louvain community detection on a similarity graph, and affinity propagation. Affinity propagation had the highest average intra-cluster coherence and, on qualitative inspection, produced more semantically homogeneous clusters, so we used it for the analysis.

Study 2 investigated AI assistance in rule creation using a fictional WhatsApp group conversation. This controlled stimulus enabled systematic comparison between human-authored rules (with and without AI assistance), while eliminating variability from user-uploaded conversations.

5.1 Procedures for Study 2

We conducted a between-subjects experiment comparing two conditions: **Human Only Condition**, where participants wrote rules for a fictional family WhatsApp group themselves, and **Human + AI (HAI) Condition**, where participants could iteratively prompt an LLM to generate rules for the given group or edit the generated rules. We recruited participants from Prolific, who resided in India, were fluent in English, and served as a WhatsApp group admin.

To provide participants with a realistic context for rule creation, we designed a fictional family WhatsApp group featuring eight relatives engaged in typical family group interactions, exchanging around 60 messages over three days. We carefully constructed the conversation to include four issues based on LLM-generated rules in Study 1 that were highly selected by participants across different group types. These issues included:

1. **Adding an Unknown Number**: A group member adds an unknown phone number to the group without prior announcement, causing confusion among members who question the unfamiliar participant.
2. **Sharing sensitive content**: A member forwards a political message from Instagram that spirals into a heated discussion.
3. **Disrespectful language**: The heated discussion escalates to group members using profane language.
4. **Sharing unverified commercial links** A group member shares multiple product links to a suspicious website.

The conversation was presented in a realistic interface replicating WhatsApp's original style of message bubbles, timestamps, sender names, system messages (e.g., member additions/ removals), indication of deleted messages, and link previews with thumbnails (refer Figure 15). To control for potential order effects, we created three versions of the same conversation, where we shuffled the order in which different issues appear in the group. Participants were randomly assigned to one of the three versions. To ensure that participants thoroughly reviewed the conversation before writing rules, we tracked whether they scrolled to the end of the conversation before they could submit the rules. The complete procedure for Study 2 is depicted in Figure 3.

Human-Only Rule Creation Participants wrote rules for the fictional family WhatsApp group by themselves. We provided a simple text interface and instructed them to write as many rules as they wished, formatted as a list. To ensure substantive engagement, we required participants to write at least 20 words. We also disabled the paste feature so that participants could not copy from external content.

HAI Collaborative Rule Creation Participants collaborated with an LLM[2] to create rules for the same fictional family WhatsApp group. The interface included two primary components: a prompt area where participants could describe what rules they wanted the AI to generate, and the output area where AI-generated rules appeared and could be edited. The fictional WhatsApp conversation was provided to the model in a structured JSON format along with the user's prompt to enable full context rule generation. We only appended an extra instruction to 'not add unnecessary text' and to 'write succinctly' to make fair comparisons with human-only written rules.

Participants could prompt the AI for rule generation as many times as they wanted. However, each interaction replaced the previously generated rules, but participants could freely edit the generated rules, delete unwanted sections, or submit the generated rules unchanged.

Data Collection Participants in both conditions answered basic demographic questions (e.g., age, gender, district and state of residence, education level) and WhatsApp usage patterns (usage frequency, number of groups where the participant serves as an admin, adminship period). Additionally, we collected four *self-reported* attitudinal measures on a 5-point likert scale around rule-creation experience derived from previous work on AI-assisted writing and ideation (Zhang et al. 2023; Kim et al. 2023). These are:

- **Confidence** that the rules are appropriate for the group.
- Level of **Control** participants felt when writing the rules.
- **Ease** of writing the rules.
- **Satisfaction** with the final rules.

Rule Evaluation The central hypothesis of Study 2 is that group rules written with AI assistance are perceived as higher quality than those written by humans alone. For this, we conducted crowd evaluations where participants rated a pair of randomly selected human-written rule set and AI-assisted rule set across the following dimensions:

1. **Clarity**: if rules are easy to read and unambiguous since policies can exceed typical reading levels (Nahrgang et al. 2025; Benoliel and Becher 2019).
2. **Contextual fit**: if rules align with group norms because governance is context-dependent (Chandrasekharan et al. 2018).
3. **Enforceability**: if rules can be applied consistently in practice as they often do not cleanly translate into actions (Jhaver et al. 2019; Jiang et al. 2019).
4. **Perceived legitimacy**: if rules appear fair and acceptable to group members because procedural justice shapes compliance (Tyler 2006; Katsaros et al. 2022).
5. **Tone**: if rules use socially appropriate wording for a family WhatsApp group.

Each participant rated 10 randomly sampled pairs, where each set of rules appeared only in one pair in a randomized

[2]We used Gemini 2.5 Pro with with a temperature of 1.0 and top-p sampling of 0.9 to encourage diverse and creative rule suggestions

order (check Figure 18). For crowd evaluation, we initially recruited the same participants from Prolific who had written the rules after reading the fictional family group conversation since they were already familiar with the rule creation context. However, we ensured that participants only rated rules created by others instead of their own. We also conducted evaluations by a separate group of raters recruited via Clickworker, who had no prior context about these rules. Information for instructions and compensation for writing rules and evaluation is provided in screenshots Figures 20 and 21 respectively.

5.2 Study 2: Findings

We recruited $N = 55$ participants for writing rules after appropriate power analysis (described in appendix C.1), with 20 assigned to the human-only condition and 35 to the HAI condition using Bernoulli Randomization (Imbens and Rubin 2015). Both conditions were well-balanced on observed covariates: age ($M = 33.5$, $SD = 11.4$), gender (67.3% male), education (54.5% bachelor's, 36.4% master's), WhatsApp usage frequency (92.7% daily users), and administrative experience ($M = 3.8$ years, $SD = 1.9$). All standardized mean differences were below 0.25, and no between-group comparisons reached statistical significance (see Table 6 in the appendix for details).

Writing Experience We examined whether AI assistance affected participants' subjective experience of the rule-writing task. Participants in both conditions reported high levels of satisfaction, ease, confidence, and perceived control (median = "Agree" or "Strongly Agree" across all measures), with no significant differences between conditions (all $p > .45$; ordinal regression). However, in the human-only condition, 70% of participants "Strongly Agreed" that they were confident in their rules, compared to only 43% in the HAI condition. This pattern is notable given the subsequent crowdsourced evaluations show that AI-assisted rules were consistently perceived as superior.

We also measured how much participants in the HAI condition ($N = 35$) edited the suggested rules by computing text similarity between the AI-generated version and the final submitted version.[3] Most participants (65.7%, n=23) made no edits at all and one participant (2.9%) made only minor tweaks. The rest made more meaningful changes: 14.3% (n=5) made moderate edits like rewording or small deletions while keeping the structure, and 17.1% (n=6) made major revisions such as substantial rewrites, adding new rules, or reorganizing the entire set. We found little evidence that the high no-edit rate was primarily driven by over-reliance on AI. In a post-hoc behavioral analysis (Appendix C.2), most non-editors—17 of 23 (74%)—showed substantive engagement across multiple measures, including writing prompts and time spent on task. This suggests that, for many participants, non-editing may have reflected satisfaction with the generated rules rather than passive acceptance.

Perceived Rule Quality To test whether AI-assisted rules were perceived as higher quality, we fit Bradley-Terry models predicting the probability whether the AI-assisted rule set would be chosen over the human-only rule set in each pairwise comparison. Table 2 presents the results.

Table 2: Bradley-Terry model estimates for pairwise rule comparisons. All comparisons are between-condition (human-only vs. human + AI). β is the log-odds advantage of AI-assisted rules; OR is the odds ratio; All p-values survive Holm correction at $\alpha = .05$.

Metric	n_{decisive}	β	SE	OR	PS	p
Clarity	476	0.549	0.090	1.73	.634	$< .001$
Contextual Fit	438	0.716	0.092	2.05	.672	$< .001$
Tone	432	0.557	0.090	1.75	.636	$< .001$
Enforceability	462	0.665	0.092	1.94	.660	$< .001$
Perceived Legitimacy	459	0.627	0.091	1.87	.652	$< .001$

In crowd evaluation, AI-assisted rules were preferred over human-only rules across all five dimensions. The largest effect was observed for Contextual Fit (OR $= 2.05$, $PS = .672$), indicating that raters found AI-assisted rules more than twice as likely to fit for the group's context. Effects for Enforceability (OR $= 1.94$) and Perceived Legitimacy (OR $= 1.87$) were similarly strong. Even the smallest effect: Clarity (OR $= 1.73$) indicated that AI-assisted rules were preferred nearly two-thirds of the time. All effects were highly significant ($p < 10^{-9}$) and robust to Holm correction for multiple comparisons.

As a robustness check, we re-estimated the main effect using logistic regression with cluster-robust standard errors (clustering by rater). This approach also yielded substantively identical conclusions (see Appendix Table 7).

Rater Condition Effects Because the same participants who wrote the rules also served as raters for rule pairs, we examined whether a rater's writing condition influenced their evaluations. This analysis addresses a potential concern: do participants who used AI assistance simply rate AI-assisted rules more favorably because they recognize rules similar to the ones they had submitted?

We extended the logistic regression model to include rater's writing condition as a covariate, with cluster-robust standard errors. Table 3 summarizes the estimated probability of choosing the AI-assisted rule, stratified by the rater's writing condition.

Critically, raters from *both* writing conditions preferred AI-assisted rules at rates significantly above chance. Raters who used AI assistance for writing rules showed stronger preference for AI-assisted rules (68-76% across metrics). As shown in previous studies, seeing high-quality examples can shift evaluative standards upward (Sadler 2009). Even those who had written rules themselves without AI assistance still chose AI-assisted rules 60-63% of the time across several evaluation metrics (all $p < .03$ vs. 50%, except Clarity at $p = .12$ and Perceived Legitimacy at $p = .10$). This demon-

[3] We used Ratcliff/Obershelp string matching (via Python's difflib.SequenceMatcher), then grouped edits into four mutually exclusive levels based on similarity s: no edit ($s \geq 0.99$), minor ($0.95 \leq s < 0.99$), moderate ($0.80 \leq s < 0.95$), and major ($s < 0.80$).

Table 3: Probability of choosing the AI-assisted rule by rater's own experimental condition. Estimates from logistic regression with cluster-robust standard errors, controlling for comparison order. The "Difference" column reports the p-value for the rater condition coefficient.

	Human-Only Raters		Human + AI Raters		
Metric	PS	95% CI	PS	95% CI	p_{diff}
Clarity	.599	[.473, .714]	.683	[.618, .741]	.221
Contextual Fit	.632	[.539, .717]	.757	[.703, .805]	.014
Tone	.618	[.514, .713]	.702	[.637, .761]	.157
Enforceability	.621	[.520, .713]	.730	[.664, .787]	.058
Perceived Legitimacy	.606	[.480, .719]	.717	[.645, .780]	.108

strates that the preference for AI-assisted rules is not driven solely by familiarity or preference for one's written rules.

These results were consistent when estimated via generalized linear mixed models with random intercepts for raters (Appendix Table 8) and ordinal cumulative link mixed models that retained tie responses (Appendix Table 9).

To further address potential familiarity bias, we conducted a follow-up evaluation with $N = 83$ independent raters recruited via Clickworker (after standard quality filtering; see Appendix C.3). These raters did not contribute any rules to the comparison pool and reviewed the mock WhatsApp conversation for the first time. Analysis using the Bradley–Terry model revealed greater preference for AI-assisted rules with scores ranging from 0.62–0.68 across the five dimensions (similar to scores from original rule writers 0.63–0.67), with all comparisons highly significant ($p < 10^{-9}$). Clickworker raters' preference for AI-assisted rules were closer to those of the human-only writer-rater in Table 3 ($PS = 0.60$–0.63) than to the AI-assisted subgroup ($PS = 0.68$–0.76), supporting the conclusion that the preference for AI-assisted rules is not driven by familiarity bias. Full replication results are in Appendix Table 11.

Linguistic Properties of Rules We quantitatively analyzed the rules participants wrote with or without AI.

Verbosity Rules in HAI condition were significantly longer (contained 72% more words) than the ones from human-only condition (Mann-Whitney $U = 168$, $p = .002$). The mean word count for AI-assisted rules was $M = 95.7$ ($SD = 53.1$, $Mdn = 90$) compared to $M = 55.6$ ($SD = 31.3$, $Mdn = 49$) for human-written rules.

Lexical Diversity We computed type token ration (TTR; unique word types divided by total word tokens) after removing stylistic markers (e.g., bullets, emphasis markers), converting all text to lowercase, and tokenizing only alphabetic words. When comparing rule sets, TTR was similar for human-written rules (M=0.786, SD=0.105) and AI-assisted rules (M=0.778, SD=0.080), with no significant difference (U=361, p=.713). At the corpus level, raw TTR was lower for AI-assisted text (0.163 vs. 0.333) but this was driven by a much larger AI corpus (3,253 vs. 1,113 words). Using bootstrap size-matching (1,000 samples of 1,113 words per condition), corpus-level TTR was essentially identical: human-written M=0.254 (95% CI [0.242, 0.266]) vs. AI-assisted M=0.257 (95% CI [0.242, 0.271]), *indicating no meaningful lexical differences* once length is controlled.

Topic Coverage We assessed the topics covered by rules using keyword-based topic detection. Exact details of the method is provided in appendix C.4
AI-assisted rules covered significantly diverse governance topics ($M = 9.1$, $SD = 1.8$) than human-written rules ($M = 5.2$, $SD = 2.2$) (Mann-Whitney $U = 65$, $p < .001$). On average, AI-assisted rules addressed nearly twice as many governance issues reported in Fiesler et al. (2018).

Table 12 in Appendix presents topic-level coverage rates. Three topics showed the largest differences: *privacy* (+66.2 pp), *media sharing* (+55.6 pp), and *new member protocols* (+48.2 pp). These topics were addressed by the majority of the participants in HAI condition (70–91%) but only a minority of participants in human-only condition (15–40%).

Notably, two topics featured in the fictional group conversation—*politics* and *spam*—were addressed by participants in both conditions at comparably high rates (80–100%), suggesting that content related issues are captured regardless of AI assistance.

Framing To examine whether AI-assisted rules exhibited more restrictive framing, we classified linguistic markers following (Fiesler et al. 2018). Restrictive markers limit or forbid actions (e.g., "no," "avoid," "refrain," "limit"), while prescriptive markers express guidelines (e.g., "be respectful," "inform," "maintain") (complete list is in appendix C.5). We computed the proportion of restrictive markers per participant. Both conditions exhibited predominantly restrictive framing for the provided fictional WhatsApp chat (Manual: 71.9%; AI-Assisted: 64.3%), with no significant difference between conditions (U = 382.5, p = .448).

Task Completion Time We computed task completion time as the duration between the timestamp when participants began the rule-writing task and when they submitted their final rules. The distribution of completion times is presented in figure 17 There was no significant difference in task duration between conditions: manual participants spent $M = 13.55$ minutes ($SD = 7.76$, $Mdn = 11.19$) on the task, while AI-assisted participants spent $M = 13.22$ minutes ($SD = 10.02$, $Mdn = 10.60$). The difference was not statistically significant (Mann-Whitney $U = 379$, $p = .490$). Despite the AI-assisted condition involving additional steps (writing a prompt, waiting for generation, potentially editing), participants in this condition did not take longer to complete the task. One interpretation is that the time saved by not having to generate rules from scratch offset the overhead of interacting with the AI system.

Overall, Study 2 establishes that AI assistance can lead to group rules that admins perceive as higher quality across multiple dimensions and that cover a more diverse set of governance issues.

6 Discussion

Governance with Privacy? Our investigation into AI-assisted governance on WhatsApp reveals that usable, high quality rule generation require integrating group context, which is in tension with the privacy guarantees of end-to-end encrypted platforms. However, recently WhatsApp has introduced "Private Processing" technology to offer AI-generated summaries of the group chat to individual members (Meta 2025b). This confidential processing in a virtual environment neither stores group data nor makes them available to the platform or any third party (Meta 2025a). This process can be adapted to support group admins in creating contextual rules for their groups in a privacy-preserving way.

Additionally, when provided with rationale, participants preferred generic rules as much as contextual ones. This suggests that with appropriate explanation, generic rules—requiring only the group type and no specific group context—can be equally appealing to group admins. Therefore, in a new group which does not have any activity yet or groups where admins and members are uncomfortable sharing conversation data for rule generation, WhatsApp could suggest generic rules accompanied by rationale. The platform could also let admins choose how many generic versus contextual rules they want and let admins decide how much group conversation to use for rule generation (e.g., from the past week or month), giving users greater control and stronger sense of privacy.

Although Study 1 examines one-time rule generation, in practice rules in online communities evolve as group composition and activities change over time. Platforms could allow admins to choose how often (e.g., every 2–3 months) rules are reviewed and updated based on ongoing group activity. The model could also learn from the rules that admins do not adopt. Our qualitative analysis ("Reason for Rule Selection") indicates rejection itself is informative. Participants ignored rules that were already implicitly followed, in conflict with existing group norms, or irrelevant to the group's dynamics. Each distinction potentially carries feedback about what the model should adjust in future updates. Treating these rejection as structured feedback, in the spirit of adaptive self-governance (Ostrom 1990), opens a concrete direction for follow-up work on adaptive rule creation.

AI as a Collaborative Governance Partner. Our Study 2 provides compelling evidence that AI is not merely a tool for efficiency, but also improves the perceived quality of governance rules. Participants who wrote rules themselves also rated AI-assisted rules higher on contextual fit, enforceability, and tone (perceived legitimacy and clarity were directionally consistent but did not reach significance for this subgroup). Creating rules that balance authority with approachability, anticipate edge cases, and remain enforceable–is a form of care labor that volunteer admins find exhausting (Wohn 2019; Matias 2019). LLMs may help admins navigate this load, producing sets of rules that cover a wide range of governance topics (Section "Topic Coverage") and that admins themselves perceive as higher quality on multiple dimensions.

This relates to the broader literature on AI-mediated communication, which suggests that AI can help users articulate intent more effectively than they can on their own (Buschek, Zürn, and Eiband 2021; Gero, Liu, and Chilton 2022). However, unlike creative writing tasks where voice is personal, rule writing is functional. The high scores for *Enforceability* and *Perceived Legitimacy* in our results for AI-assisted rules indicate that LLMs are particularly adept at adopting the administrative voice: a neutral, objective tone that is often difficult for users embedded in close-knit social circles to adopt without sounding authoritarian.

Furthermore, the strong performance of the AI in the *Contextual Fit* metric suggests that such features could be particularly valuable for the Global South contexts where our study was situated. In regions like India, where WhatsApp serves as essential infrastructure for everything from neighborhood watch groups to extended family networks, the diversity of group norms is immense. A one-size-fits-all set of "Community Guidelines" is insufficient. Our approach enables hyper-local governance, allowing each micro-community to establish norms that reflect its specific cultural and social reality.

Limitations and Future Work. Our study has several limitations. First, we recruited participants only from India, WhatsApp's largest market. Since cultural norms around governance vary globally, future work should replicate these studies in diverse geographies to disentangle cultural preferences around AI-assisted rule generation.

Second, Study 2 evaluated the content of generated rules rather than downstream outcomes such as compliance, conflict reduction, or group health. Still, perceived legitimacy is a strong predictor of compliance (Tyler 2006; Katsaros et al. 2022), and admin's perception of rule quality is a prerequisite for adoption. A natural next step would be a field deployment following the CivilServant methodology proposed by Matias (2019), randomly assigning AI-assisted and human-only rule sets across opt-in WhatsApp groups and measuring outcomes such as conflict levels, member churn, admin interventions, and rule violations.

Third, our post-hoc analyses cannot fully rule out automation bias, nor can the experimental data confirm whether participants carefully evaluated accepted AI-generated rules. We observed that two participants in the HAI condition exerted minimal effort in prompt writing and accepted rules unconditionally. The long-term effects of blindly adopting AI-generated rules therefore remain an open question for future research. Fourth, we used a single model (Gemini-2.5) for rule generation, and some observed tendencies—such as the prevalence of prescriptive rules—may be model-specific. Future work should compare the capabilities and outputs of different models.

Finally, users' trust in the privacy protections of E2EE platforms remains mixed (Fischer et al. 2025). Consequently, it is unclear whether groups would adopt contextual rules even if designed to preserve privacy, though deployment-side safeguards such as client-side timestamp coarsening and session-level pseudonym rerandomization may further reduce residual risks. Future fieldwork should examine real-world adoption and community perceptions of such tools.

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Ethics checklist

1. For most authors...
   (a) Would answering this research question advance science without violating social contracts, such as violating privacy norms, perpetuating unfair profiling, exacerbating the socio-economic divide, or implying disrespect to societies or cultures? Yes
   (b) Do your main claims in the abstract and introduction accurately reflect the paper's contributions and scope? Yes
   (c) Do you clarify how the proposed methodological approach is appropriate for the claims made? Yes
   (d) Do you clarify what are possible artifacts in the data used, given population-specific distributions? Yes
   (e) Did you describe the limitations of your work? Yes
   (f) Did you discuss any potential negative societal impacts of your work? Yes. We discuss the risk of automation bias and over-reliance on AI-generated rules in the Limitations section.
   (g) Did you discuss any potential misuse of your work? Yes. We discuss the risk of admins blindly adopting AI-generated rules without scrutiny in the Limitations section.
   (h) Did you describe steps taken to prevent or mitigate potential negative outcomes of the research, such as data and model documentation, data anonymization, responsible release, access control, and the reproducibility of findings? Yes
   (i) Have you read the ethics review guidelines and ensured that your paper conforms to them? Yes
2. Additionally, if your study involves hypotheses testing...
   (a) Did you clearly state the assumptions underlying all theoretical results? Yes
   (b) Have you provided justifications for all theoretical results? NA
   (c) Did you discuss competing hypotheses or theories that might challenge or complement your theoretical results? No
   (d) Have you considered alternative mechanisms or explanations that might account for the same outcomes observed in your study? Yes
   (e) Did you address potential biases or limitations in your theoretical framework? NA
   (f) Have you related your theoretical results to the existing literature in social science? Yes
   (g) Did you discuss the implications of your theoretical results for policy, practice, or further research in the social science domain? Yes
3. Additionally, if you are including theoretical proofs...
   (a) Did you state the full set of assumptions of all theoretical results? NA
   (b) Did you include complete proofs of all theoretical results? NA
4. Additionally, if you ran machine learning experiments...
   (a) Did you include the code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL)? NA
   (b) Did you specify all the training details (e.g., data splits, hyperparameters, how they were chosen)? NA
   (c) Did you report error bars (e.g., with respect to the random seed after running experiments multiple times)? NA
   (d) Did you include the total amount of compute and the type of resources used (e.g., type of GPUs, internal cluster, or cloud provider)? NA
   (e) Do you justify how the proposed evaluation is sufficient and appropriate to the claims made? NA
   (f) Do you discuss what is "the cost" of misclassification and fault (in)tolerance? NA
5. Additionally, if you are using existing assets (e.g., code, data, models) or curating/releasing new assets, **without compromising anonymity**...
   (a) If your work uses existing assets, did you cite the creators? Yes
   (b) Did you mention the license of the assets? No. It is unpublished
   (c) Did you include any new assets in the supplemental material or as a URL? No
   (d) Did you discuss whether and how consent was obtained from people whose data you're using/curating? Yes. We discuss our IRB-approved data-donation protocol and the limitations of partial (donor-only) consent in the Privacy and Ethics paragraph of Section 3.
   (e) Did you discuss whether the data you are using/curating contains personally identifiable information or offensive content? Yes
   (f) If you are curating or releasing new datasets, did you discuss how you intend to make your datasets FAIR (see Wilkinson et al.)? NA
   (g) If you are curating or releasing new datasets, did you create a Datasheet for the Dataset (see Gebru et al.)? NA
6. Additionally, if you used crowdsourcing or conducted research with human subjects, **without compromising anonymity**...
   (a) Did you include the full text of instructions given to participants and screenshots? Yes
   (b) Did you describe any potential participant risks, with mentions of Institutional Review Board (IRB) approvals? Yes
   (c) Did you include the estimated hourly wage paid to participants and the total amount spent on participant compensation? Yes
   (d) Did you discuss how data is stored, shared, and deidentified? No. This is however, documented in the IRB.

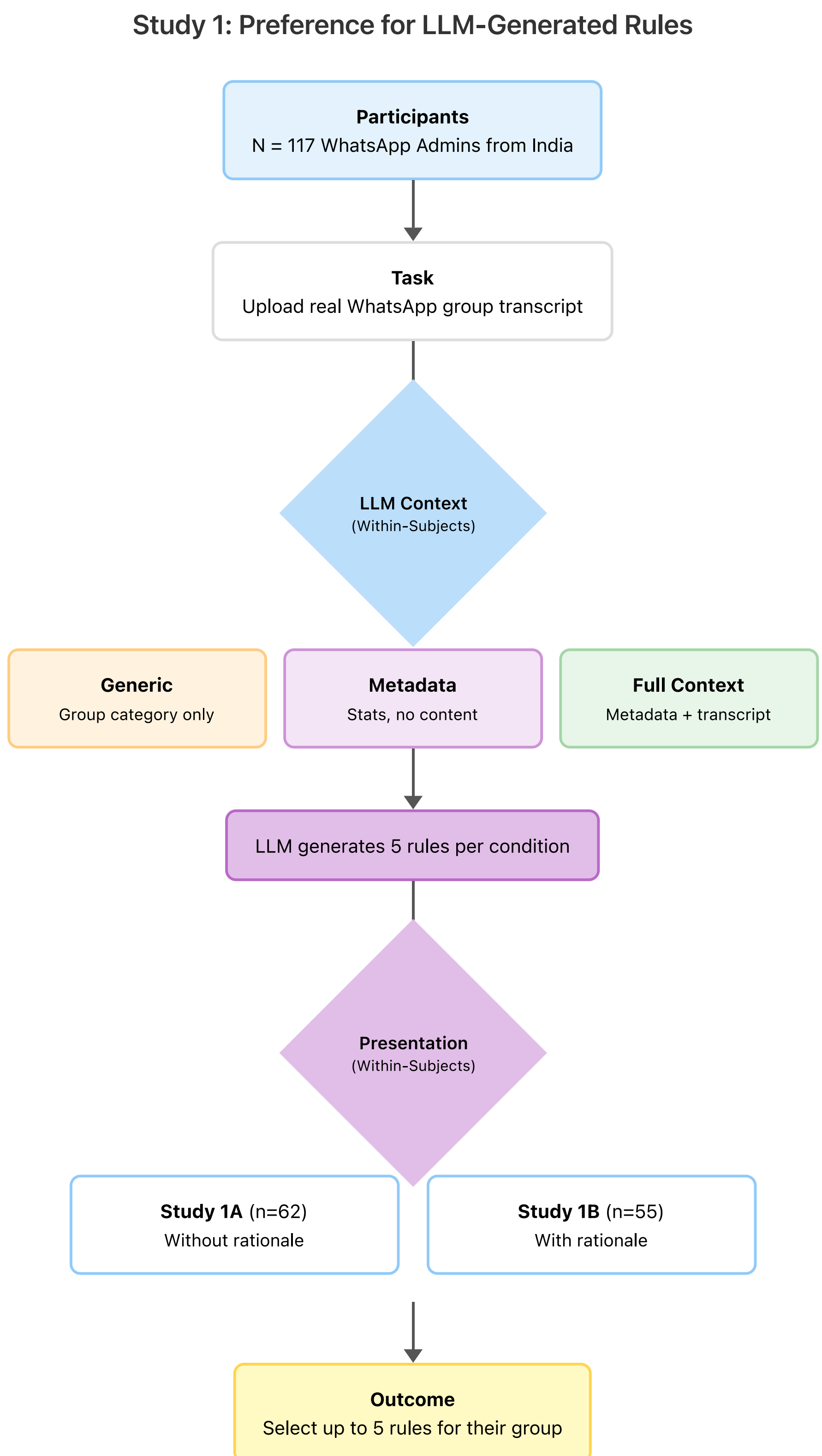


Figure 2: Flow diagram of the procedures for study 1

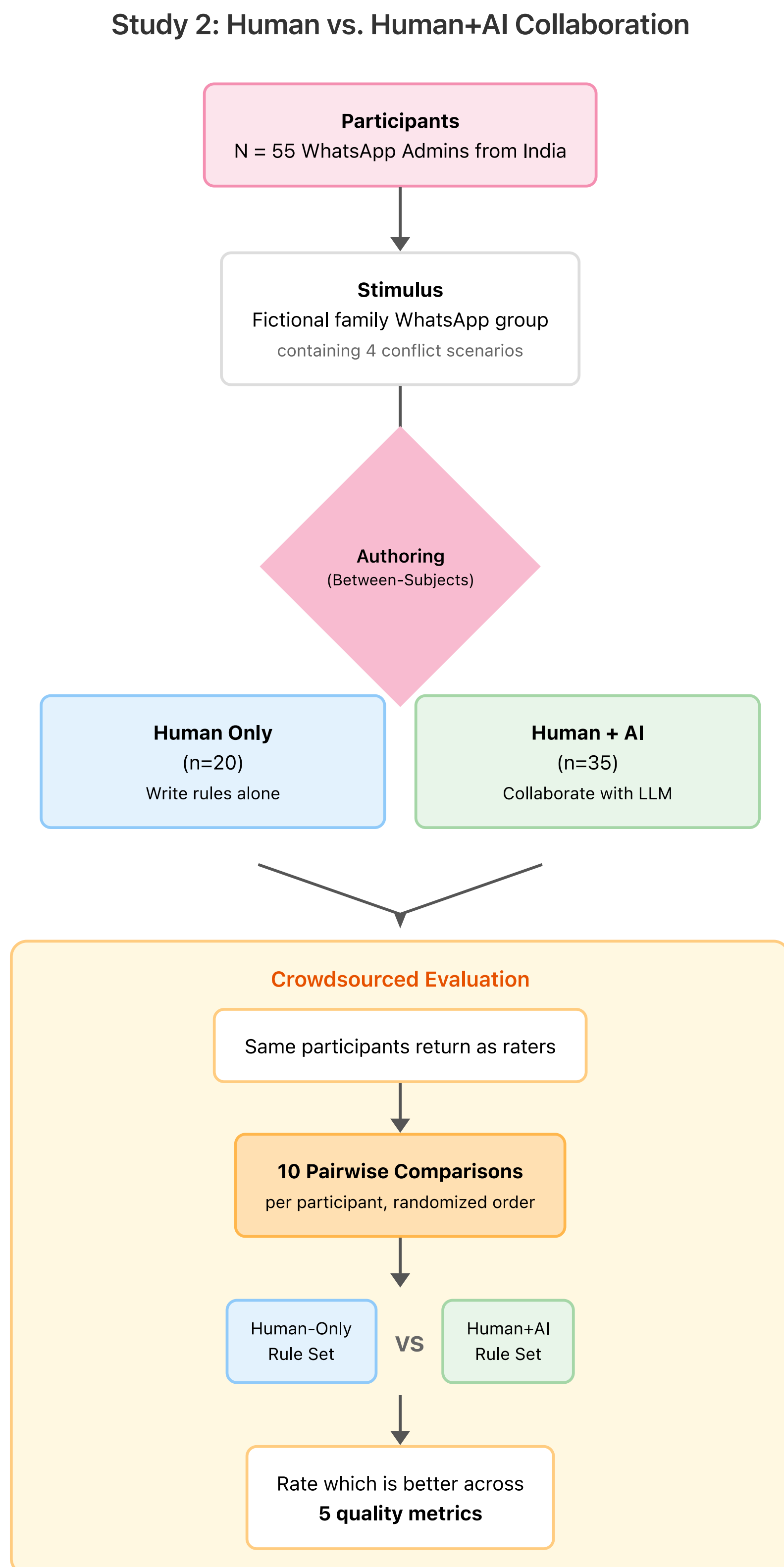


Figure 3: Flow diagram of the procedures for study 2

Table 4: Rule categorization following Fiesler et al. (2018); Leibmann et al. (2025). N=16 unique rule descriptions across 20 groups (0.027% of all groups).

Dimension	Category	Count
Tone	Restrictive (prohibits behaviors)	15 (75%)
	Prescriptive (encourages conduct)	5 (25%)
Target	Post content	14 (70%)
	User behavior	5 (25%)
	User participation	1 (5%)
Topic[†]	Off-topic / relevance	9 rules
	Enforcement / consequences	4 rules
	Spam / low-quality	3 rules
	Links / external restrictions	3 rules
	Civility / respect	2 rules
	Privacy*; externally referred norms*; other	5 rules

[†]Topics are non-exclusive (rules may address multiple topics).
*Not in Fiesler et al. (2018); Leibmann et al. (2025); added for WhatsApp-specific rules.

Appendix A: WhatsApp rule cateogories

Appendix B: Study 1

B.1 Power Analysis (Study 1)

We conducted a pilot study with 16 participants to estimate effect sizes for power analysis. For each participant, we computed the proportion of selected rules from each generation type (contextual, metadata, generic), then derived *within-subject difference scores* comparing contextual to generic and contextual to metadata proportions. Using the pilot-based mean differences and standard deviations of these within-subject contrasts, we performed a power analysis for paired t-tests (via the noncentral t distribution; see, e.g., (Cohen 1988)) with a Bonferroni-adjusted significance level for the two primary pairwise comparisons. This analysis indicated that we would require approximately $N \approx 55$ participants to achieve around 80% power to detect the observed difference between contextual and generic rule selection proportions in the main study. Power curves are shown in Figure 4

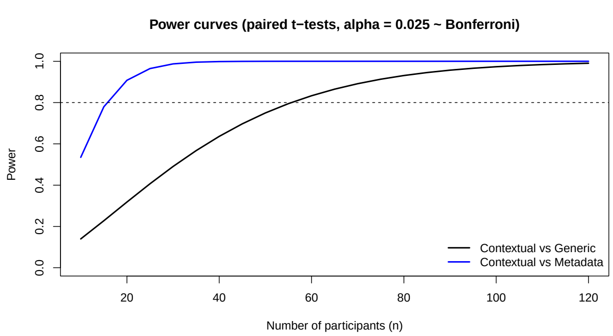


Figure 4: Power curves for Study 1

B.2 Prompt Templates for Study 1

This appendix provides the complete, verbatim prompts used for rule generation and merging in Studies 1A and 1B. Variable placeholders are shown in `${variable}` notation; these were replaced with actual values at runtime.

1. Generic Condition Prompt

```
Your task is to suggest rules for guiding
user behavior and activities
in WhatsApp groups.

Task: Provide exactly five distinct rules
that would be suitable for a
WhatsApp group categorised as
"${groupType}". Each rule must be a
single, self-contained statement (one short
self-contained sentence)
describing the expectation or restriction.
For each rule, also provide
a short "reason" sentence and a "reasoning"
explanation.

Requirements:
- Try to add a mix of prescriptive rule
(what members should do) and
  restrictive rule (what members must
  avoid).
- Focus on broadly applicable group norms
that you think would be
  applicable for "${groupType}" type, from
  your own prior knowledge.
- For each rule include a neutral, general
"reason" that explains the
  benefit of the rule without implying
  knowledge of any specific or
  anedotal knowledge of any particular
  group, transcript, or
  participants.
- Keep the tone neutral and reasoning
abstract (i.e., reasons should
  read like universal justifications, not
  case-specific commentary).
- For each rule, also provide a "reasoning"
field (2-3 sentences) that
  explains the underlying governance
  principle or logic that makes
  this rule valuable for "${groupType}"
  groups. What information that
  you have made you think that this rule is
  important?
- Keep each rule concise and actionable;
avoid duplicates and
  numbering.
- CRITICAL: Ensure all five rules are
SUBSTANTIVELY DIFFERENT from
  each other. Do NOT generate near-similar
  rules that could be easily
  merged. Verify each rule addresses a
  distinct concern.

Return JSON only in this schema:
{
  "rules": [
    {
```

```
      "text": "Rule statement",
      "reason": "Neutral, general
      justification",
      "reasoning": "Detailed explanation of
      why you thought this rule
                    is valuable (2-3
                    sentences) based on the
                    information you have"
    }
  ]
}
```

2. Metadata-Only Condition Prompt

```
Your task is to suggest rules for guiding
user behavior and activities
in WhatsApp groups.

Context summary (no message content
provided):
- Group type: ${groupType}
- Observation window: ${stats.windowStart}
to ${stats.windowEnd}
- Total messages analysed:
${stats.totalMessages}
- Participants observed
(${stats.topParticipants.length} of
  ${stats.distinctParticipants}):
${participantLines}
- Media breakdown: ${mediaBreakdown}
- Notable activity notes:
${stats.activityNotes}
- Token budget note:
${trimmingNote}

Below is an excerpt where each row is JSON
with timestamp, sender,
mediaType, and a content object that
contains only metadata (word and
character counts for
text/link/system/deleted messages; and
mediaType
plus filename when available for media
messages). No raw message text
or URLs are included.
${messagesBlock}

Task: Create exactly five distinct rules
based only on these metadata
signals that you think are the most relevant
or important. Express
each rule as a single, self-contained
statement (one short
self-contained sentence) that sets clear
expectations or boundaries
for the group. For each rule, also provide a
short "reason" sentence
and a "reasoning" explanation.

Requirements:
- Do not infer or reference any specific
message content, quotes, or
  topics. Provide rules solely based on
  activity patterns and
  metadata. You can use the evidence in the
  metadata or infer user
  patterns based on the metadata for which
  you can make rules.
- Mix of prescriptive and restrictive rules.
- Keep the tone constructive and neutral;
avoid naming individuals or
  exposing any personal data.
- Reasons should be phrased in neutral
general terms and sound generic
  in tone and abstraction. The rules
  themselves should still be
  strictly drawn from the metadata
  information of the chat
  transcripts.
- For each rule, also provide a "reasoning"
field (2-3 sentences) with
  specific evidence or inference logic that
  you used from the provided
  metadata that led to this rule. What
  activity metdata
  information/signals support this
  recommendation? Include specific
  observations you found in the metadata as
  your reasoning.
- Avoid numbering, bullet symbols, or extra
commentary.

Return JSON only in this schema:
{
  "rules": [
    {
      "text": "Rule statement",
      "reason": "Neutral, general
      justification grounded in metadata",
      "reasoning": "Specific inference
      logic/ evidence from the
                    information/signals from
                    the transcript metadata
                    that led to this rule
                    (2-3 sentences with
                    examples)"
    }
  ]
}
```

3. Full Contextual Condition Prompt

```
Your task is to suggest rules for guiding
user behavior and activities
in WhatsApp groups.

Context summary:
- Group type: ${groupType}
- Observation window: ${stats.windowStart}
to ${stats.windowEnd}
- Total messages analysed:
${stats.totalMessages}
- Participants observed
(${stats.topParticipants.length} of
  ${stats.distinctParticipants}):
${participantLines}
- Media breakdown: ${mediaBreakdown}
- Notable activity notes:
${stats.activityNotes}
- Token budget note:
${trimmingNote}
```

```
Below is an excerpt containing the most
recent ${messages.length}
group activities and messages from the
selected window. Each row is
JSON with timestamp, sender, mediaType
(text|image|video|audio|
document|link|system), and content. Use it
extensively to understand
recurring topics, conflicts, and norms.
${messagesBlock}

Task: Create exactly five distinct rules
tailored to the observed
group behaviours. Express each rule as a
single, self-contained
statement (one short self-contained
sentence) that sets clear
expectations or boundaries for the group.
For each rule, also provide
a short "reason" sentence and a "reasoning"
explanation.

Requirements:
- Each rule must be grounded in patterns and
in the transcript, but do
  not mention the transcript, chat logs,
  participants, or the analysis
  process. Do not write "this group" or
  otherwise reveal that these
  rules come from a specific dataset.
- Mix of prescriptive and restrictive rules.
- Reference specific behaviours only when
they appear in the excerpt,
  but phrase the accompanying "reason" in
  neutral, general and
  abstract justification rather than
  explicit references to the
  observed messages.
- Keep the tone constructive and neutral;
avoid naming individuals or
  exposing personal data.
- Keep reasons generic enough in tone and
level of abstraction. The
  rules themselves should still be grounded
  from observations in the
  transcript.
- For each rule, also provide a "reasoning"
field (2-3 sentences) with
  specific evidence or inference that you
  made from the provided
  transcript that made you suggest this
  rule. What patterns,
  behaviors, or observations in the messages
  support this
  recommendation? Reference specific
  examples or inference logic that
  made you recommend this rule.
- Avoid numbering, bullet symbols, or extra
commentary.

Return JSON only in this schema:
{
  "rules": [
    {
      "text": "Rule statement",
      "reason": "Neutral, general
      justification grounded in patterns",
      "reasoning": "Specific evidence or
      inference logic based on the
                     provided transcript with
                     examples that led to
                     this
                     rule (2-3 sentences
                     only)"
    }
  ]
}
```

B.3 Threat Model and Residual Leakage

Our three conditions sit on a spectrum from no group data (generic) to full message content (full-context). The metadata condition is the middle ground, giving the model behavioral signals without sending the actual messages. When we claim it offers "better privacy protection," we mean better than sending the full transcript, not zero leakage. The residuals are explicit below.

Prompts are sent to Google Cloud AI Gemini (paid tier), which does not use our data for training and retains logs only briefly for abuse detection. HTTPS protects transmission.

The *generic* condition sends only the group category (e.g., "Family"). No participant-level information leaves the device.

The *metadata* condition sends the group type, observation window, total messages, per-participant pseudonym and activity counts, media-type breakdown, a weekly hour-of-day, volume summary aggregated across the group, and per-message JSON rows with timestamp, sender pseudonym, mediaType, and word/character counts (filenames for media). It does not send message text. What still leaks: timestamps tied to the same pseudonyms across messages reveal when each member is active and who *possibly* replies to whom; counts and media ratios reveal group composition and culture; message-length distributions reveal the shape of conversations; and filenames might carry personal information (names, dates, places).

The *full-context* condition adds the full message text after on-device redaction of URLs, emails, and phone-like numbers. What this does not catch: things people mention in passing —health issues, family disputes, political or religious views, locations, names of people not in the group — none of which look like structured PII. Stylometric matching against external writing samples is also a risk for any individual whose other writing is available.

Some of these residuals can be reduced further in deployment with additional client-side processing: stripping filenames, coarsening timestamps to the hour, rerandomizing pseudonyms per session. We treat this as ongoing pipeline work.

We do not claim formal anonymization guarantees. The pipeline is best-effort: no differential privacy, no k-anonymity, no formal bound on re-identification risk against a determined adversary with side-channel information. This is the same partial-guarantee position taken across the WhatsApp data-donation literature (Garimella and Chauchard 2025; Ohme and Araujo 2022).

4. Rule Merging Prompt The merging prompt below was called once per participant after rule generation, with the 15 generated rules (5 per condition) and their rationales as input. It returned a deduplicated set in which rules from different conditions could be combined when their underlying reasoning was essentially identical (e.g., both flagged the same observed pattern in the transcript), while rules with distinct justifications were retained as separate items even if their surface text was similar. Same-source merges were forbidden: two rules from the generic condition will never be collapsed into one, regardless of textual similarity.

```
You will be given a list of rules along with
their reasoning
(evidence/inference) for a real WhatsApp
group generated from 3
different context types (Just assume these
are 3 different sources).
Some rules may be near-duplicates while
others are distinct.

Goal: Merge rules ONLY if they have very
similar reasoning (same
evidence/data OR very similar inference from
slightly different data).
Keep rules separate if reasoning differs,
even if the rule text is
similar.

Guidance for merging based on "reasoning"
similarity:
- Merge if: The reasoning uses the exact
same evidence/data OR makes
  very similar inferences about user
  behavior from slightly different
  data points.
- Do NOT merge if: The reasoning is based on
different evidence,
  different observations, or different
  behavioral patterns - even if
  the rule text looks similar.
- CRITICAL: NEVER merge rules from the SAME
source (e.g., do NOT merge
  "generic-1" with "generic-2"). ONLY merge
  rules from DIFFERENT
  sources (e.g., "generic-1" with
  "contextual-2" with "metadata-3" or
  any of those combinations).
- When merging: Pick the clearest rule text
and randomly select one of
  the similar reasoning statements (since
  they're similar, any one is
  representative).
- CRITICAL: Two rules with similar text but
different reasoning should
  remain as SEPARATE rules.

Input rules (one JSON per line):
${payloadLines}

Return JSON only in this schema:
{
  "rules": [
    {
      "text": "Canonical merged rule text",
      "reasoning": "One of the similar
      reasoning statements (one picked
                    randomly from the
                    provided)",
      "from": ["generic-1", "contextual-3",
      "metadata-2"]
    }
  ]
}
```

Worked Examples of Merging Two examples illustrate the merging step. A *two-way merge*: a generic rule "Avoid sharing forwarded messages without verifying their source and content" (general anti-misinformation principle) and a contextual rule "Verify forwarded news articles before reposting, particularly during election periods" (citing transcript instances of disputed forwards) make the same inference from slightly different evidence, so they merge into a single rule tagged `[generic, contextual]`; a participant's selection of this merged rule contributes a count to both conditions. A *three-way merge* similarly collapses generic, metadata-only, and contextual rules about off-topic content—with general-etiquette, frequency-pattern, and observed-interruption justifications respectively—into one `[generic, contextual, metadata]`-tagged rule.

Notes on Prompt Implementation Variable substitutions in the prompts above were performed at runtime as follows:

- `${groupType}`: User-selected group category (Family, Friends, Work, etc.)
- `${stats.windowStart}`, `${stats.windowEnd}`: ISO 8601 timestamps of first and last messages
- `${stats.totalMessages}`: Total message count in transcript
- `${stats.topParticipants.length}`: Number of top contributors shown
- `${stats.distinctParticipants}`: Total unique participants
- `${participantLines}`: Formatted list of top participants with message counts
- `${mediaBreakdown}`: Comma-separated counts by media type
- `${stats.activityNotes}`: Heuristic description of activity patterns
- `${trimmingNote}`: Description of whether older messages were trimmed for token budget and how many messages were dropped
- `${messagesBlock}`: Newline-separated JSON objects (one per message) containing timestamp, sender pseudonym, media type, and either full content (contextual) or metadata only (metadata-only)
- `${messages.length}`: Number of messages included after trimming
- `${payloadLines}`: Newline-separated JSON objects for each rule including id, source, text, and reasoning

All prompts were sent to Google's Gemini 2.5 Pro API (`models/gemini-2.5-pro`) with temperature 0.65 and top-P 0.9, except the merging prompt which used Gemini 2.5 Flash (`models/gemini-2.5-flash`) with identical generation parameters.

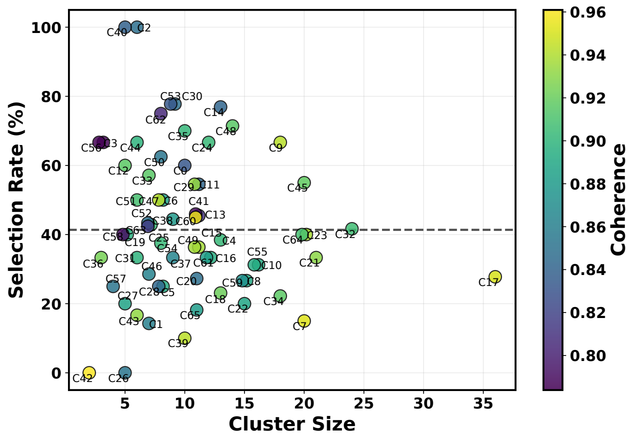


Figure 5: Cluster size vs. Selection Rate for Exp 1A

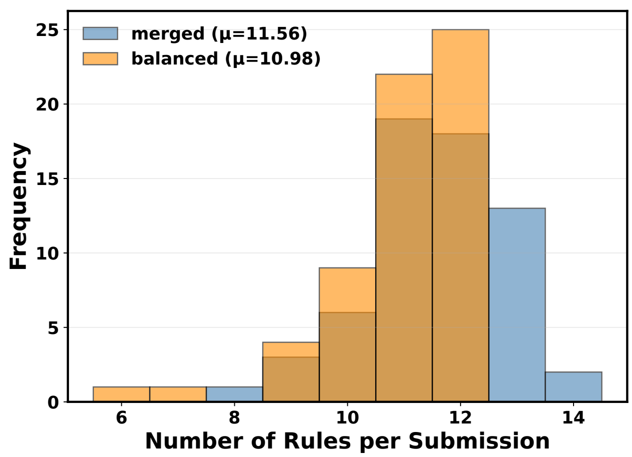


Figure 6: Distribution of total rules after merging vs. after dropping for study 1A

B.4 Rule Set Dropping and Balancing via Constrained Randomization

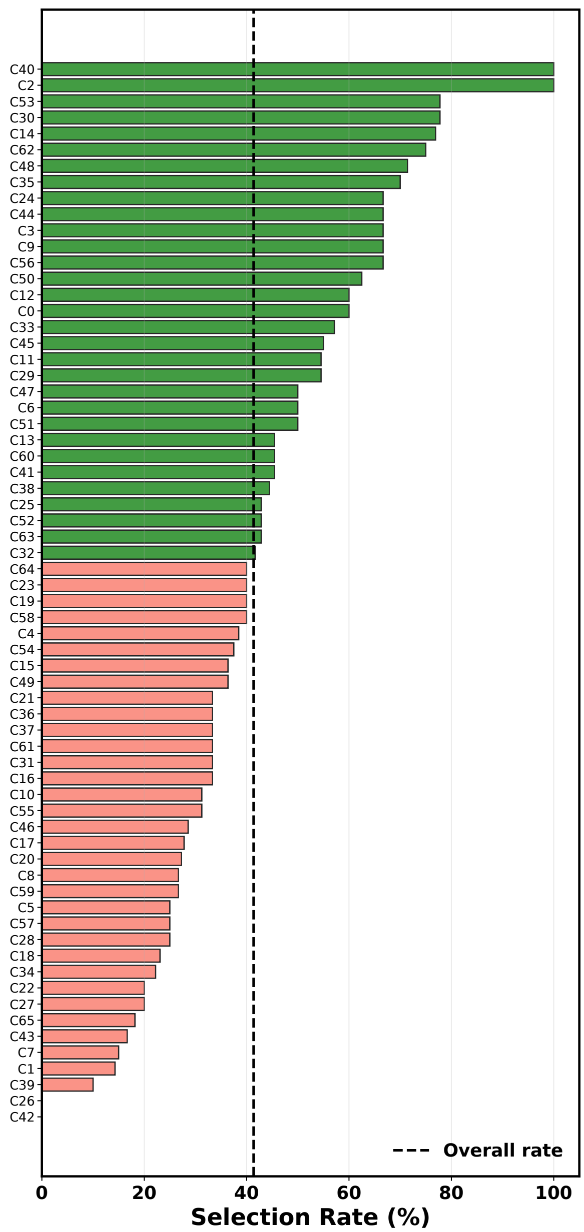
Following the merging stage, the rule set typically contained between 10 and 15 rules, depending on how many merges occurred. However, we faced a methodological challenge: if we presented all merged rules to participants, the relative representation of the three information conditions would be unbalanced. For instance, if the generic condition's rules were highly distinctive while the metadata and contextual conditions generated similar rules that merged together, participants would see more pure-generic rules than pure-metadata rules, potentially biasing the analysis toward generic rules simply because of their numerical dominance in the choice set.

To address this imbalance while avoiding arbitrary rule selection, we developed a constrained randomization balancing algorithm. The algorithm's goal was to reduce the rule set to exactly 12 rules (our target based on pilot testing of cognitive load—12 rules could be meaningfully compared without overwhelming participants) while maintaining equal representation from all three source categories. Representation was counted inclusively: if a merged rule carried tags `[generic, contextual]`, it contributed one count to generic and one count to contextual.

The balancing algorithm operated iteratively with a maximum of three iterations (sufficient to achieve balance in 95% of cases based on pilot data). At each iteration, the algorithm computed the current count of rules from each source category. If the counts were already balanced and the total was 12 or fewer, the algorithm terminated. Otherwise, it identified which categories were over-represented relative to the minimum count (e.g., if counts were {`generic: 5, metadata: 3, contextual: 4`}, metadata would be the minimum at 3, so generic and contextual would be over-target). The algorithm then evaluated each rule to determine whether dropping it was a valid move—that is, whether there existed a feasible path to a balanced state within the remaining iterations if that rule were removed.

This feasibility check employed a lookahead mechanism: for each candidate rule, the algorithm simulated removing it, updated the counts, and ran a greedy simulation to determine the minimum number of additional drops needed to reach balance. If that minimum was less than or equal to the number of iterations remaining, the rule was considered a valid candidate. This lookahead approach prevented the algorithm from making locally attractive drops that would later create unresolvable imbalances. For example, if dropping a rule that was tagged `[generic, metadata]` would leave generic over-represented with no remaining rules that could be dropped to fix it, that rule would not be a valid candidate despite being multi-sourced.

Once the set of valid candidates was identified, the algorithm randomly selected one rule from this set with uniform probability, using a seeded random number generator initialized with the participant's session identifier. This seeding ensured reproducibility—the same participant would always see the same balanced rule set if they restarted the session—while randomizing across participants to prevent systematic biases in which types of rules were dropped. The selected rule was removed, the counts were updated, and the algorithm proceeded to the next iteration. The algorithm terminated when either a balanced state of 12 or fewer rules was achieved or the maximum iteration count was reached. In practice, nearly all cases converged within three iterations; the few cases that did not were retained with their unbalanced counts rather than forcing further drops, since over-dropping would undermine the experimental manipulation by reducing the choice diversity.



Cluster	Themes (top-5)	G/C/M (%)
C40	language, common, use, ensure, members	0/100/0
C2	adhere, deadlines, strictly, assignments, tasks	0/100/0
C53	opportunities, job, relevant, professional, share relevant	0/100/0
C30	banter, jokes, personal, avoid, respectful	0/100/0
C14	availability, confirm, scheduled, event, hours	0/100/0
C62	verify, information, sharing, report, requests	25/75/12
C48	professional purpose, group professional, purpose, professional, strictly	100/57/57
C35	respectful, tone, maintain, interactions, maintain respectful	70/30/0
C24	personal attacks, attacks, personal, refrain personal, refrain	67/67/0
C44	divisive, refrain, topics, refrain sharing, debates	50/50/0
C3	debate, member, restrict, administrative, actions	0/100/0
C9	unverified, chain, chain messages, unverified information, forwarding	100/6/6
C56	report, task, reporting, send, designated	0/100/0
C50	expenses, shared, settle, activities, agreed	0/100/0
C12	communicate clearly, concisely, clearly concisely, clearly, communicate	80/20/40
C0	acknowledge, receipt, promptly, direct, direct mentions	10/90/10
C33	let, conversation let, space, contribute, members contribute	0/0/100
C45	advertisements, commercial, unsolicited, refrain, unsolicited advertiseme	65/45/0
C11	pinned, asking, question, review, recent	0/100/0
C29	learning topic, learning, topic, discussions shared, discussions	91/91/82
C47	non work, work related, work, forwards, related media	88/12/0
C6	deleting, messages, context, conversation, avoid	0/12/100
C51	resources, articles, share, relevant, notes	67/33/0
C13	clearly, deadlines, state, location, details	36/64/9
C60	use mention, mention feature, feature direct, mention, direct questions	82/18/9
C41	job, channel, details, requests, post	9/100/9
C38	feedback, constructive, offer, help, provide	67/22/11
C25	images, professional, media, relevant group, directly relevant	29/14/100
C52	share, images, media, share images, relevant	0/43/71
C63	event, polls, group, specific, time location	0/100/0
C32	relevant, purpose, group, shared, shared content	79/50/38
C64	brief, images, description, brief text, sharing	0/15/95
C23	engage, ideas, respectful, engage respectful, engage discussions	95/10/0
C19	continue, achievements, personal, share personal, share	0/100/0
C58	time, scheduled, message, include, time location	0/100/0
C4	members encouraged, encouraged, members, contribute, encouraged con	0/0/100
C54	contribute, questions, ask, actively, asking	75/12/25
C15	consolidate multiple, consolidate, multiple, album, single message	9/73/91
C49	review, review messages, sending, deletions, sending minimize	9/9/100
C21	share, group, information, externally, confidential	100/5/0
C36	complex, scheduled, topics, discussions, moving	0/33/100
C37	private, conversations, private messages, private chat, private message	44/44/22
C61	reply, use reply, respond specific, specific messages, use	42/50/17
C31	concise, discussions, focused, important, purpose	33/67/67
C16	consolidate, single, thoughts, single message, multiple	8/17/100
C10	mindful message, mindful, timing, hours, message frequency	56/38/88
C55	contribute, members, space, encourage, members contribute	0/12/88
C46	refrain, sending, refrain sending, greetings, short	43/57/0
C17	thoughts single, thoughts, single, consolidate, short	22/14/100
C20	direct, administrative, designated, contact, questions	0/100/0
C8	forwards memes, avoid, memes, volume, forwards	73/20/27
C59	simple, instead sending, acknowledgements, use, instead	27/47/40
C5	allow, contribute, allow space, space, sending multiple	12/0/100
C57	information, shared, sharing, verify, proprietary	50/50/0
C28	clearly, facilitate, context, provide, message	62/38/0
C18	adding new, adding, new, new members, group adding	77/23/0
C34	standard, hours, standard business, urgent, business hours	89/67/89
C22	personal information, information, personal, explicit, consent	73/27/0
C27	readability, longer, improve, lengthy, consider	0/0/100
C65	sharing multiple, multiple, album, multiple photos, photos	0/45/82
C43	provide, links, provide brief, description, sharing links	0/33/100
C7	late, night, late night, morning, early morning	70/50/100
C1	disagreements, address, respectfully, chat, private	43/57/0
C39	late night, late, night, mindful posting, hours	10/10/100
C26	announcements, lengthy, post, share, readability	0/80/20
C42	sensitive, provide, content, sharing, sharing media	50/50/0

Figure 7: Themes across clusters and composition of different context conditions for Study 1A. Cluster composition from different data contexts (generic, contextual, metadata) is presented in the right table.

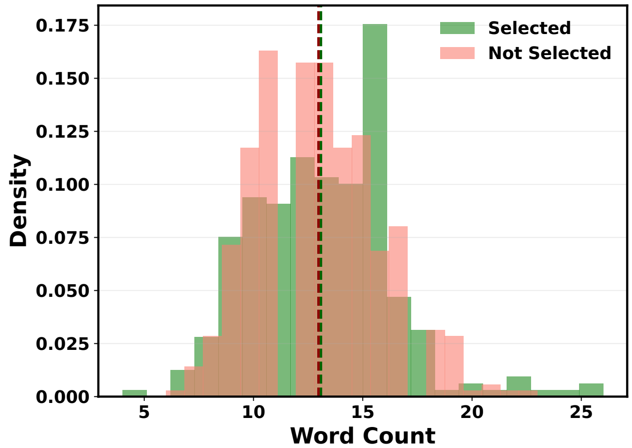


Figure 8: Distribution of word counts across selected and non-selected rules in Study 1A

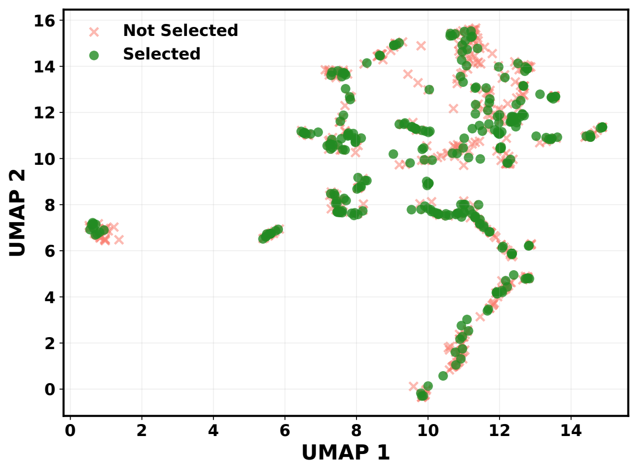


Figure 9: UMAP visualization of rule embeddings for Study 1A

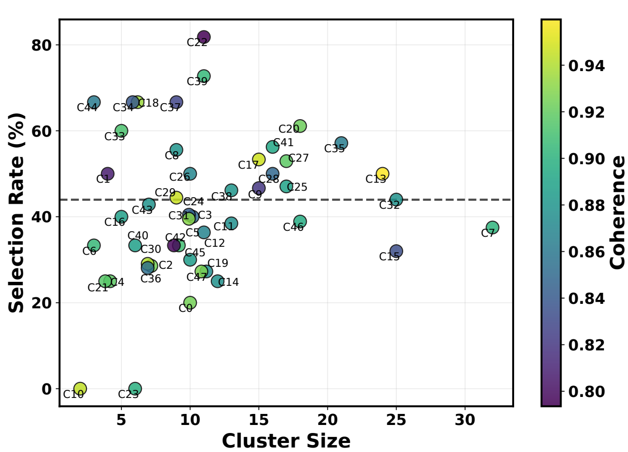


Figure 10: Cluster size vs. Selection Rate for Study 1B

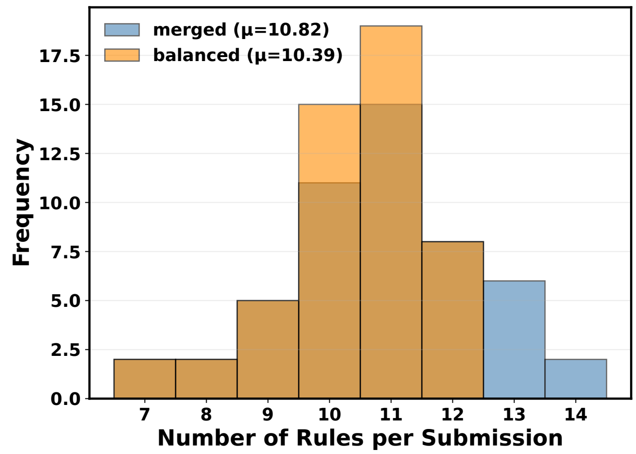


Figure 11: Distribution of total rules after merging vs. after dropping for Study 1B

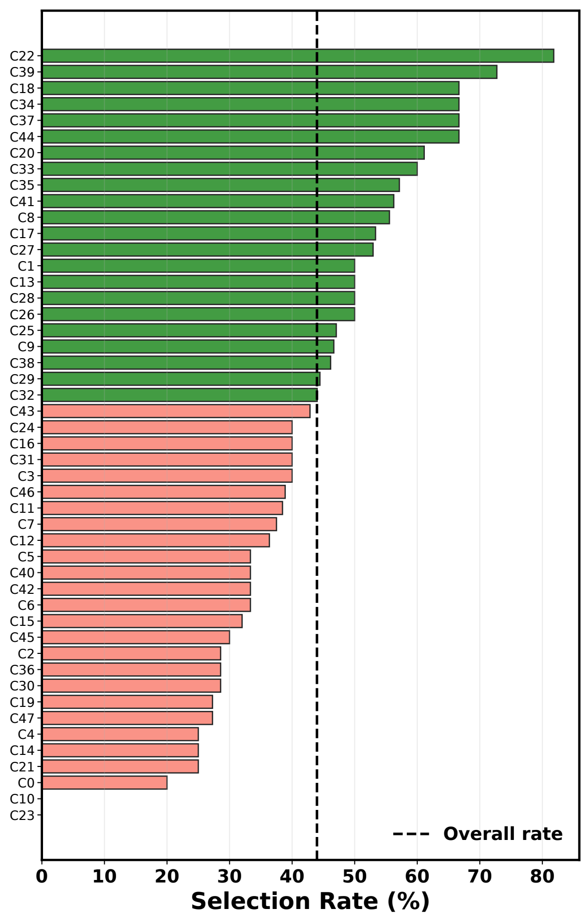
Worked Example of the Algorithm. The algorithm iteratively drops rules (maximum 3 iterations). At each iteration, it admits a rule as a candidate only if dropping it preserves a feasible path to a balanced state of ≤ 12 unique rules within the remaining iterations—this is the lookahead. It then picks uniformly at random from the admitted candidates using a session-seeded RNG. Two cases illustrate the choice space.

Case 1 (two-way merge). The merged set contains 4 pure-generic, 4 pure-contextual, 5 pure-metadata rules, and one merged rule X tagged $\{g, c\}$—14 unique rules, inclusive counts $5/5/5$. Two end states are reachable. (i) Drop X, then drop one uniformly-sampled pure-metadata rule: 12 unique pure rules at $4/4/4$. (ii) Drop one pure-generic, one pure-contextual, and one pure-metadata rule: 11 unique rules including X, also at $4/4/4$.

Case 2 (three-way merge). The merged set contains 4 pure-generic, 4 pure-contextual, 4 pure-metadata rules, and one merged rule Y tagged $\{g, c, m\}$—13 unique rules, inclusive counts $5/5/5$. Two end states are reachable. (i) Drop Y alone: 12 unique pure rules at $4/4/4$, terminating in one iteration. (ii) Drop one pure rule from each condition: 10 unique rules including Y, at $4/4/4$.

In both cases, which end state is realized depends on the participant's session seed; the sensitivity analysis (Appendix B.5) simulates this counterfactual variation by sampling uniformly over the full enumeration of valid end states.

This balancing approach provided several methodological advantages. First, it ensured that no single information condition dominated the choice set, enabling fair comparison of selection rates across conditions. Second, the randomization introduced variation in which specific rules were dropped across participants, preventing any single rule from disproportionately influencing aggregate results. Third, the seeded randomization maintained within-participant reproducibility while ensuring between-participant variation, which is important for future work examining how individual differences moderate rule preferences.



Cluster	Themes (top-5)	G/C/M (%)
C22	clear, decisions, event, group, activities	0/91/9
C39	acknowledgements, instead sending, simple, use, instead	27/64/82
C18	information, confidential, sensitive, share, information group	83/17/0
C34	sources, share, unverified, information, news	50/50/17
C37	location, requests, help, offers, clear	44/78/0
C44	time, specific, initiating, date, planning	0/100/0
C20	engage, engage discussions, respectfully, opinions, especially	89/22/0
C33	adding new, adding, new, group, member	80/20/0
C35	professional, professional purpose, directly relevant, relevant group, dire	76/62/48
C41	use reply, reply, reply feature, use, reply function	31/38/38
C8	check, question, asking, requesting, materials	22/89/0
C17	screenshots, share screenshots, share, non members, group non	100/0/0
C27	standard, hours, hours respect, personal time, time	88/59/82
C1	assigned, confirm, action, provide brief, item	0/100/0
C13	short, thoughts single, thoughts, single, message instead	46/21/96
C28	respectful, professional, personal, tone, positive	31/75/0
C26	let, space, contribute, mindful, space members	0/10/90
C25	purpose, group, relevant, purpose group, discussions	76/35/29
C9	advance, plans, confirm, changes, availability	0/100/0
C38	treat, confidential, group confidential, treat information, shared group	77/23/0
C29	review, review messages, minimize need, sending minimize, deletions	0/0/100
C32	chain messages, chain, unverified, news, unverified news	84/36/8
C43	feedback, offer, constructive, focus, advice	86/29/0
C24	academic, conversations focused, focused, conversations, related	40/100/20
C16	new members, new, members, group, adding new	80/20/0
C31	advertisements, commercial, refrain posting, posting, refrain	50/50/0
C3	deleting, follow message, message, follow, deleting messages	20/20/60
C46	text description, brief text, description, brief, text	0/39/94
C11	multiple, single, album, message album, consolidate	8/69/100
C7	late, late night, night, messages late, night early	56/38/91
C12	single, consolidate, multiple, consolidate related, related thoughts	45/82/73
C5	divisive, highly divisive, topics unless, initiating, highly	89/22/0
C40	specific, specific individuals, mention feature, mention, individuals	50/33/17
C42	clear, announcements, include, required, location	11/89/0
C6	period, avoid, opportunity, avoid posting, work	33/67/0
C15	direct, main group, group, private, main	32/60/28
C45	brief, relevance, include brief, include, sharing link	10/60/70
C2	follow, allow, respond, allow space, participation	0/0/100
C36	share, contribute, resources, projects, seek	57/43/14
C30	deleting, deleting messages, conversation refrain, refrain deleting, conve	0/0/100
C19	encourage, members, contribute, questions, members contribute	0/18/100
C47	album, sharing multiple, photos, sharing, multiple photos	0/55/91
C4	acknowledge, different, immediate, responses, availability	50/25/25
C14	consolidate, reactions, single, messages, use reactions	42/75/75
C21	civic, objectives, relevant group, aid, directly relevant	75/50/75
C0	address, disagreements, address personal, chat address, personal disagre	70/30/0
C10	flooding chat, album, avoid flooding, flooding, consolidate	0/100/100
C23	detailed, consider, summarizing, text, lengthy	0/33/100

Figure 12: Themes across clusters and composition of different context conditions for Study 1B. Cluster composition from different data contexts (generic, contextual, metadata) is presented in the right table.

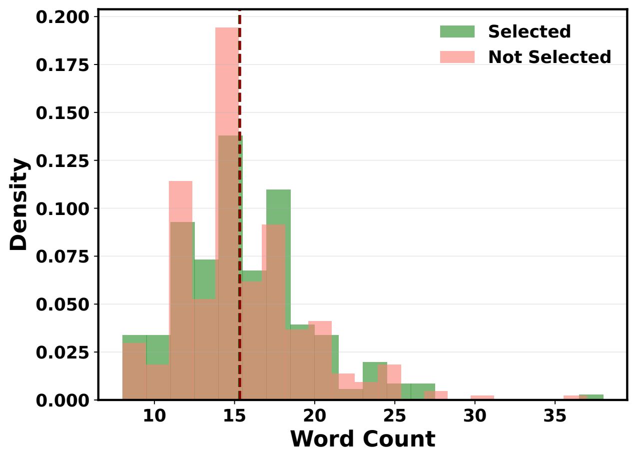


Figure 13: Distribution of word counts across selected and non-selected rules in Study 1B

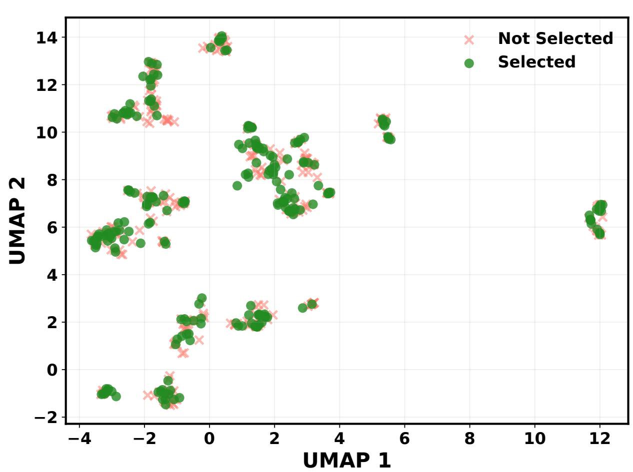


Figure 14: UMAP visualization of rule embeddings for Study 1B

B.5 Sensitivity Analysis of the Dropping Procedure

The balancing algorithm described in Appendix B.4 drops a small number of rules per participant to maintain equal per-condition representation, with the specific rules retained determined by a session-seeded pseudo-random algorithm. This raises a question: would Study 1 preference results have altered substantially if the algorithm had retained a different subset? Because we observe selections (and non-selections) only for the rules each participant actually saw, the selections for dropped rules constitute unobserved counterfactual data—fundamentally *a missing-data problem*. To assess sensitivity, we simulate alternative dropping outcomes via Monte Carlo, impute counterfactual selections under the exchangeability assumption, and combine the resulting estimates via Rubin's Rules (Rubin 1987) to obtain inference that accounts for both sampling and counterfactual uncertainty.

The dropping procedure affected only a minority of the generated rules: 42 of 62 participants in Study 1A and 43 of 55 in Study 1B had no rules dropped at all (the merged set was already ≤ 12 rules), and the median across participants who did experience drops was a single dropped rule.

Procedure: For each recorded participant i, let $\mathcal{M}_i$ denote the full original rule set after merging (the union of rules shown and rules dropped). We enumerated all valid balanced subsets $\mathcal{S} \subseteq \mathcal{M}_i$ satisfying $|\mathcal{S}| \leq 12$ and equal per-condition counts across the three conditions. This enumeration spans the full combinatorial space of rule combinations that can possibly be shown to participant i by the dropping algorithm, including configurations in which different multi-labeled rules are kept versus dropped. We then simulated $M = 2000$ counterfactual dropping outcomes by sampling $\mathcal{S}$ uniformly from this enumeration. Uniform sampling treats all reachable end states as equally likely, which is a conservative proxy for the algorithm's true (non-uniform) distribution over end states. For rules in $\mathcal{S}$ originally shown, we used the participant's actual selection information; for rules originally dropped but newly appearing in $\mathcal{S}$, we imputed a Bernoulli outcome with probability $\hat{p}_{i,r}$ equal to the participant's empirical selection rate for rules of that source condition (for multi-tagged rules, the mean across source conditions).

For each iteration m we computed the paired t-test on the resulting per-participant difference scores to obtain a point estimate $\hat{\theta}_m$ and within-iteration variance $\widehat{V}_{W,m}$. We combined the M iteration-level estimates using the multiple imputation rules of Rubin (1987). Letting $\bar{V}_W = M^{-1} \sum_m \widehat{V}_{W,m}$ denote the average within-iteration variance and $V_B = (M-1)^{-1} \sum_m (\hat{\theta}_m - \bar{\theta})^2$ the between-iteration variance, the pooled total variance is

$$T = \bar{V}_W + \left(1 + \tfrac{1}{M}\right) V_B,$$

where the factor $(1 + 1/M)$ adjusts for using a finite number of imputations. Pooled confidence intervals use the Barnard–Rubin small-sample degrees-of-freedom adjustment (Barnard and Rubin 1999). The total variance captures both sampling uncertainty (variation across participants in any single experiment) and counterfactual uncertainty (variation across alternative dropping outcomes).

Table 5 compares the observed study 1 estimates to the pooled inference. The between-iteration variance V_B is between 16 and 25 times smaller than the within-iteration variance $\bar{V}_W$ across all four comparisons, indicating that the dropping procedure contributes approximately 4–6% of total uncertainty. The Contextual vs. Metadata findings remain significant at the Bonferroni-adjusted threshold ($\alpha = .025$) under the pooled inference in both studies (1A and 1B), with the Study 1B comparison borderline ($p = .022$). The Contextual vs. Generic findings remain non-significant under both observed and pooled inference. Since $\hat{p}_{i,r}$, the participant's MLE rate, doesn't propagate uncertainty about $\hat{p}$ itself (an improper imputations in Rubin (1987)'s sense), we conducted a robustness check on the imputation procedure; we re-ran the analysis under proper Bayesian imputation, drawing $\hat{p}_{i,r}$ from a $\text{Beta}(s_{i,r}+1, k_{i,r}-s_{i,r}+1)$ posterior in each iteration rather than fixing it at the maximum-likelihood estimate; pooled p-values shifted by less than 0.002 in all four comparisons, and all qualitative conclusions held.

Table 5: Observed and pooled results for Study 1 preference tests.

	Observed		**Pooled (MC)**	
Comparison	**97.5% CI**	p	**97.5% CI**	p
Study 1A ($N = 62$)				
Ctx vs. Gen.	$[-.007, .181]$	.038	$[-.018, .172]$	.068
Ctx vs. Meta.	$[.078, .258]$	$< .001$	$[.064, .246]$	$< .001$
Study 1B ($N = 55$)				
Ctx vs. Gen.	$[-.053, .125]$	.349	$[-.060, .122]$	.431
Ctx vs. Meta.	$[.005, .198]$	.018	$[.003, .200]$	.022

Note. Ctx = contextual; Gen. = generic; Meta. = metadata-only; MC = Monte Carlo. CIs are reported at the 97.5% level to align with the Bonferroni-adjusted threshold $\alpha = .025$ for two planned comparisons per study.

Appendix C: Study 2

Appendix: Study 2 Supplementary Tables

Table 6: Balance test for Study 2. SMD = Standardized Mean Difference (Human + AI − Human-only). Values below 0.25 indicate adequate balance.

Variable	Human-Only ($n = 20$)	HAI ($n = 35$)	SMD	p
Age, M (SD)	34.8 (12.2)	32.7 (11.0)	−0.18	.517
Gender (% Male)	80.0%	60.0%	−0.20	.222
Education (% Bachelor's)	60.0%	51.4%	−0.09	.560
WhatsApp Freq. (% Daily)	95.0%	91.4%	−0.04	.741
Admin Experience (years)	3.5 (1.9)	3.9 (1.9)	0.23	.430

C.1 Power Analysis

For Study 2, we conducted a priori power analysis using pilot data from $N = 25$ raters who each completed 10 pair-

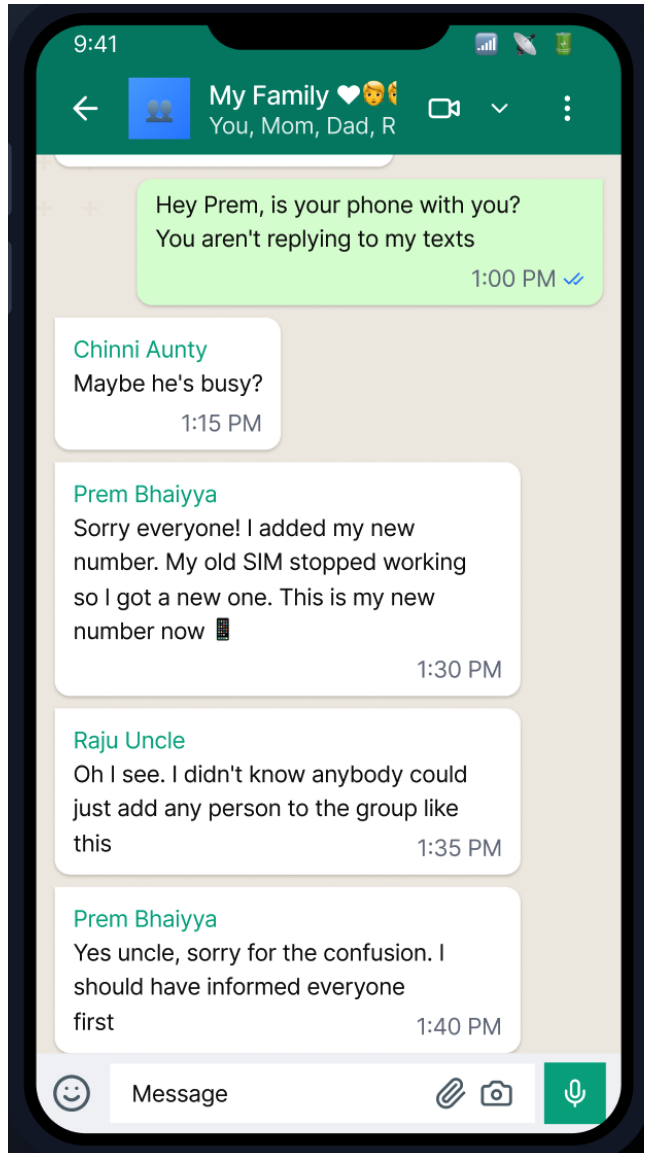


Figure 15: A screenshot of the scrollable simulated WhatsApp group chat containing problematic stimuli

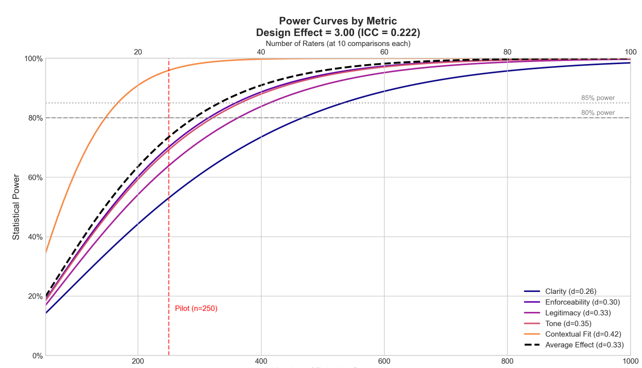


Figure 16: Power curves for crowdsourced evaluation

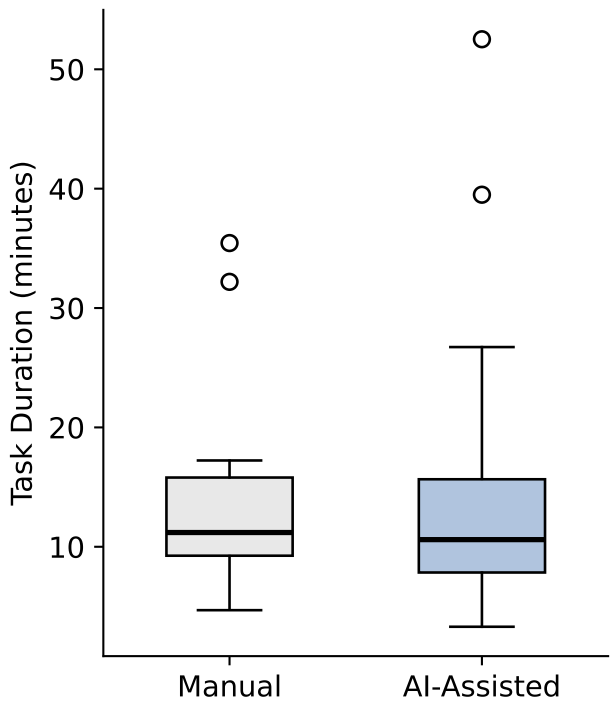


Figure 17: Box plots for the distribution of writing task completion times in Study 2

wise comparisons, yielding 250 total comparisons. The fundamental hypothesis test asks whether PS $\neq 0.5$, where

$$\mathrm{PS} = \Pr\big(\text{AI-assisted wins} \mid \text{no tie}\big) \tag{1}$$

denotes the probability of superiority–the probability that a randomly selected AI-assisted rule is judged superior to a randomly selected human-only rule. This is a one-sample binomial proportion test, for which standard closed-form power formulas exist.

Because each rater contributed multiple non-independent comparisons, we adjusted for clustering using the design effect. We estimated the intraclass correlation coefficient (ICC) from the pilot data as $\rho = 0.222$, reflecting moderate within-rater consistency. The design effect was computed as:

$$\mathrm{DEFF} = 1 + (m-1)\rho = 1 + (10-1)(0.222) = 3.00 \tag{2}$$

where $m = 10$ is the number of comparisons per rater. The effective sample size is then $n_{\text{eff}} = n/\mathrm{DEFF}$, which inflates the standard error to provide a conservative power estimate.

From the pilot data, the smallest observed effect was for Clarity (PS $= 0.611$) (figure 16). Using the normal approximation to the binomial test with the DEFF-adjusted effective sample size, we determined that detecting this effect at $\alpha = 0.05$ (two-sided) with 80% power required approximately 380 comparisons, or 38 raters at 10 comparisons each. To ensure adequate power across all five evaluation dimensions with a margin for attrition, we targeted recruitment of 55 raters for the full study.

Table 7: Primary GLM results with cluster-robust standard errors (clustering by rater). Model: logit$[PS] = \beta_0$. This directly tests whether AI-assisted rules are chosen at a rate different from chance (50%).

Metric	n	$\hat{\beta}_0$	$\mathrm{SE}_{\text{robust}}$	PS	95% CI	p
Clarity	476	0.615	0.134	.649	[.587, .707]	$< .001$
Contextual Fit	438	0.885	0.121	.708	[.657, .754]	$< .001$
Tone	432	0.693	0.126	.667	[.610, .719]	$< .001$
Enforceability	462	0.772	0.129	.684	[.627, .736]	$< .001$
Perceived Legitimacy	459	0.733	0.147	.675	[.610, .735]	$< .001$

Note. This analysis serves as a robustness check on the BT estimates in Table 2. Where BT treats each pairwise comparison as independent, this logistic regression with standard errors clustered at the rater level accounts for within-rater dependence (each rater contributes multiple comparisons).

Table 8: Generalized linear mixed model (GLMM) with random intercepts for raters, controlling for rater condition and comparison order. Model: logit$[PS] = \beta_0 + \beta_1 \cdot$ RaterCondition $+ \beta_2 \cdot$ PairIndex $+ u_{\text{rater}}$.

Metric	$\hat{\beta}_0$	$\hat{\beta}_{\text{RaterCond}}$	SE	z	p
Clarity	0.302	0.374	0.293	1.28	.202
Contextual Fit	0.491	0.607	0.237	2.56	.010
Tone	0.394	0.381	0.268	1.42	.156
Enforceability	0.316	0.555	0.271	2.04	.041
Perceived Legitimacy	0.461	0.534	0.319	1.67	.094

Note. This analysis serves as a robustness check on the rater-condition contrasts in Table 3. Where the main analysis used cluster-robust standard errors, the GLMM with random intercepts for each rater explicitly models between-rater variability in baseline preference for AI-assisted rules. Substantive conclusions about the difference between writer-rater subgroups are unchanged.

C.2 Post-hoc Analysis of Over-reliance on LLMs

To examine over-reliance on AI, we compare the 23 HAI non-editors (people who don't edit LLM generated rules after prompting) with the 12 HAI editors (people who made substantive changes to the generated rules), and with the 20 human-only participants where relevant, on three behavioral measures. We hypothesize that participants who accept AI output uncritically are less motivated to complete the task to the best of their ability and spend lesser time engaging with the tasks in Study 2 compared to the engaged participants.

Engagement Evaluation. We constructed three binary engagement signals per HAI participant: (i) prompt length $\geq$ 20 words, (ii) prompt contains at least $\geq$ 1 problematic scenario in the mock WhatsApp group, and (iii) time on the rule-writing task (from reading the conversation to writing/prompting for rules and submission) $\geq$ 8 minutes. The length and time thresholds correspond to the lower tertile (33rd percentile) of the within-HAI distribution. The topic signal captures whether the prompt addressed the substantive content of the conversation at all. For each participant's prompt, AI-generated rules, and final submitted rules, we identified whether each scenario was mentioned via case-insensitive regex matching on on a curated set of lexical stems per scenario as described in Section C.4.

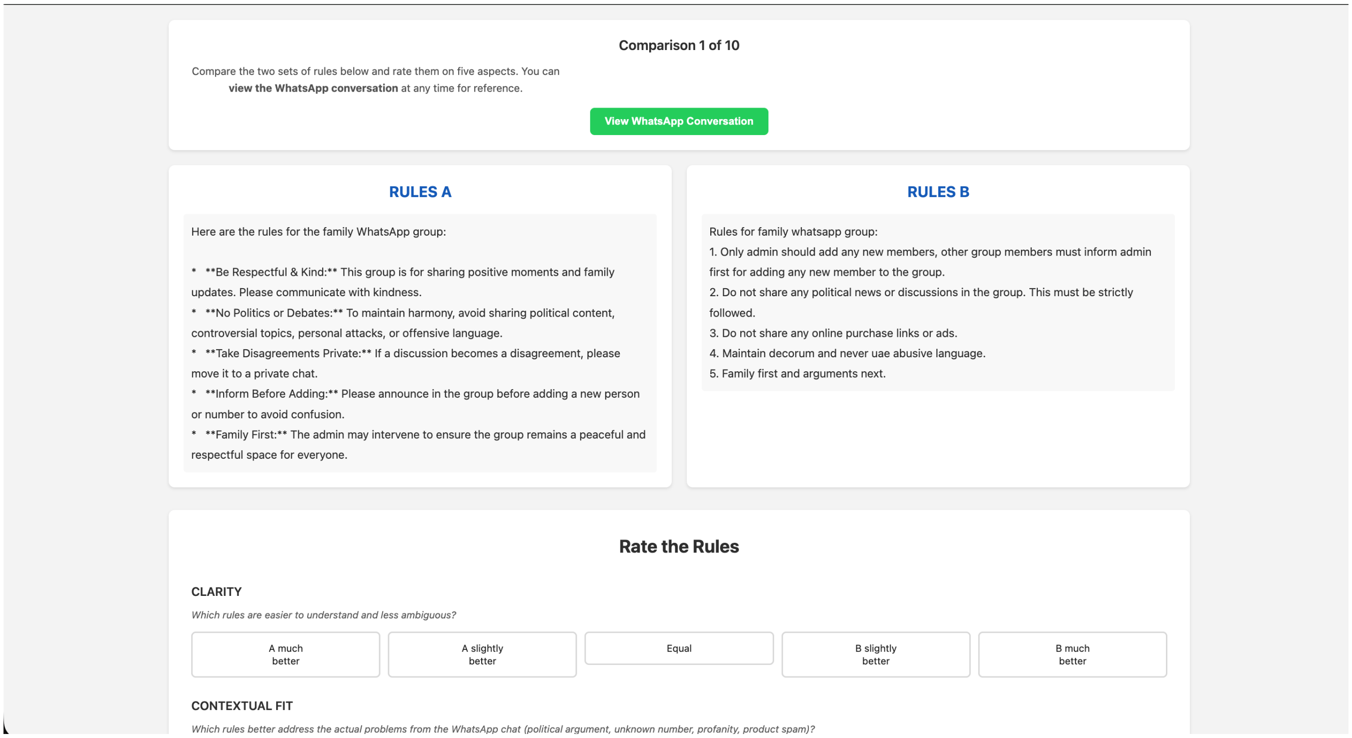


Figure 18: A screenshot of the evaluation interface for comparing human-only and human + ai generated rules for study 2

Table 9: Ordinal cumulative link mixed model (CLMM) retaining tie responses. The outcome is a 5-level ordered factor ("B much better" to "A much better"), recoded so that positive coefficients indicate preference for the AI-assisted rule.

Metric	n (incl. ties)	$\hat{\beta}_{\text{RaterCond}}$	SE	z	p
Clarity	530	0.315	0.257	1.23	.221
Contextual Fit	530	0.519	0.213	2.44	.015
Tone	530	0.251	0.208	1.21	.228
Enforceability	530	0.242	0.259	0.93	.350
Perceived Legitimacy	530	0.368	0.261	1.41	.159

Note. This analysis is a robustness check on the rater-condition contrasts in Tables 3 and 8. Where those analyses collapse the 5-point preference scale to binary and exclude "Equal" ratings as non-decisive, the cumulative link mixed model (with random rater intercepts) retains all five preference levels as an ordered outcome and keeps tied comparisons. Substantive conclusions about the rater-condition effect are unchanged.

Engagement profile and within-HAI comparisons. Most non-editors engaged substantively with the task (Table 10): 17 of 23 (74%) met $\geq$ 2 of the three signals, including 10 (43%) who met all three. Only 2/23 (9%) met none of the signals, consistent with weaker engagement. Editors showed a comparable distribution (8/12 at $\geq$ 2 signals), so the engagement pattern is not specific to the editing decision. Mann-Whitney two-sided tests with Cliff's δ on the individual measures showed no meaningful differences between groups: task time 11.4 vs. 10.0 minutes ($p = 0.50$, $\delta = +0.15$); prompt word count 28 vs. 24 ($p = 0.86$, $\delta = +0.04$); embedded scenarios in prompt median 1 vs. 1 ($p = 0.43$, $\delta = -0.16$). All effect sizes are small to negligible.

Table 10: Distribution of engagement signals met among HAI non-editors and editors. 74% of non-editors met $\geq$ 2 of 3 signals; 9% met none.

Signals met (of 3)	Non-editors ($n = 23$)	Editors ($n = 12$)
3 of 3	10 (43%)	3 (25%)
2 of 3	7 (30%)	5 (42%)
1 of 3	4 (17%)	2 (17%)
0 of 3	2 (9%)	2 (17%)
$\geq$ 2 signals	17 (74%)	8 (67%)

Qualitative examples. Two illustrations from opposite ends of the engagement distribution. *Typical strong engagement* (representative of the 17/23 non-editors meeting $\geq$ 2 signals): *"Need help writing rules for a group. In this group courtesy and politeness should be maintained. Before adding someone permission must be sought from admin and other members. No circulation of fake or abusive content."* (35 words; covers two embedded scenarios; 7.3 minutes on task). *Typical weak engagement* (representative of the 2/23 meeting zero signals): *"Please generate rules for this family whatsapp group."* (8 words; covers no specific scenario; 7.3 minutes on task).

Alignment between non-editor prompts and AI output. We cannot cleanly interpret non-editing as uncritical acceptance because most non-editors had written substantive

Purpose:

In this study, we examine if AI can help create rules for WhatsApp groups.

Study Procedure:

If you agree to participate, you will need to complete a 25-30 minute online session:

- First, you will upload a WhatsApp group chat from one of your groups. We will remove all personal information from the chat (e.g., name, phone number, address, media etc.) on your device before you upload them.
- Next, you will review some rules that will be generated by AI based on your uploaded WhatsApp group chat.
- Finally, you will answer some questions about your opinion of these rules.

Eligibility:

You must be:

- At least 18 years old,
- Living in India,
- Fluent in English,
- An active WhatsApp group admin.

Risks:

We expect minimal risk from your participation in this study. Your WhatsApp group chat will be completely anonymized on your device before you upload them. We will delete the group data after generating group rules. If you feel uncomfortable sharing your anonymous group chat, you can withdraw anytime.

Benefits

No direct benefits. Your participation potentially helps improve WhatsApp group features.

Compensation

You will receive $3.5 USD (~315 INR) via Prolific/Clickworker if you complete the study.

Confidentiality

Your identity and data are protected. We will only collect your Prolific/ Clickworker ID to process compensation. All data will be encrypted and accessed by the research team only.

Voluntary participation

Participation is voluntary. You may withdraw anytime by closing your browser. At this moment, we can only compensate those who complete the whole study.

Figure 19: Consent page for study 1

Purpose:

In this study, we examine if AI can help create rules for WhatsApp groups.

Study Procedure:

If you agree to participate, you will need to complete a 15-20 minute online session:

- First, you will review a WhatsApp group conversation.
- Next, you will write rules for this group (with or without AI assistance).
- Finally, you will answer some questions about yourself.

Eligibility:

You must be:

- At least 18 years old,
- Living in India,
- Fluent in English,
- An active WhatsApp group admin.

Risks:

We expect minimal risk from your participation in this study. The WhatsApp conversation you will review is entirely fictional and anonymized. If you feel uncomfortable at any time, you can withdraw.

Benefits

No direct benefits. Your participation potentially helps improve WhatsApp group features.

Compensation

You will receive $3.5 USD (~315 INR) via Prolific/Clickworker if you complete the study.

Confidentiality

Your identity and data are protected. We will only collect your Prolific/ Clickworker ID to process compensation. All data will be encrypted and accessed by the research team only.

Voluntary participation

Participation is voluntary. You may withdraw anytime by closing your browser. At this moment, we can only compensate those who complete the whole study.

Contact

Figure 20: Consent page for writing rules in study 2

Purpose:

In this study, we examine the quality of rules created for WhatsApp groups with and without AI assistance.

Study Procedure:

If you agree to participate, you will need to complete an 8-12 minute online task:

In this task, you will see 10 pairs of rules written by other participants for the same WhatsApp family group chat you read earlier.

For each pair, you'll compare them on five aspects:

- **Clarity:** Which rules are easier to understand and less ambiguous?
- **Contextual Fit:** Which rules better address the actual problems that occurred in the chat (the political argument, the unknown number, profanity, product spam)?
- **Tone:** Are the rules written in a style appropriate for a family group?
- **Enforceability:** As a group admin, which rules would make it easier to identify violations and take action?
- **Perceived Legitimacy:** Which rules would you be more willing to follow if they were posted in your family WhatsApp group?

Note: Neither set of rules is your own. Choose based on your honest judgment. It's okay to select "Equal" if you genuinely can't decide.

Eligibility:

You must have completed the previous WhatsApp rule-writing study to participate in this follow-up task.

Risks:

We expect minimal risk from your participation in this study. If you feel uncomfortable at any time, you can withdraw.

Benefits

No direct benefits. Your participation potentially helps improve WhatsApp group features.

Compensation

You will receive $2.00 USD (~180 INR) via Prolific/Clickworker if you complete the study.

Confidentiality

Your identity and data are protected. We will only collect your Prolific/Clickworker ID to process compensation. All data will be encrypted and accessed by the research team only.

Voluntary participation

Participation is voluntary. You may withdraw anytime by closing your browser. At this moment, we can only compensate

Figure 21: Consent page for comparing rules in study 2

prompts and then received outputs that addressed those prompts. Among the 17 non-editors who referenced at least one embedded scenario, 16 received AI output covering every scenario they requested, and only 1 received partial coverage. We see from Table 10 that a substantive number of non-editors had higher engagement signals compared to the editors. A natural hypothesis is that non-editors did not have to edit after rule generation because they had substantively engaged with the prompt upfront and were satisfied with their rules, compared to the editors who did not provide enough information upfront. This pattern is more consistent with appropriate trust in output that matched participants' stated needs than with blind acceptance. However, we do not observe the counterfactual: whether these participants would still have accepted the rules, or edited them, had the LLM failed to address what they asked for. We still acknowledge that 2/23 non-editors showed very little engagement on the composite, and we do not claim the data uniformly absolves the no-edit decision; the aggregate pattern is supported by the remaining 21.

C.3 Independent-Rater Replication of Study 2

To address potential familiarity bias from using rule-writers as raters, we recruited an additional pool of new evaluators via Clickworker. These raters had not contributed any rules to the comparison pool and read the embedded conversation for the first time. Each evaluated up to 10 randomly assigned rule pairs (one rule from the human-only condition and one from the AI-assisted condition per pair) on the same five dimensions as Study 2 using the same 5-point preference scale.

Quality Filters. Of 108 raw responses, we excluded entries failing any of the following pre-registered checks: (i) failed attention check (7); (ii) duplicate Clickworker `participantId` (2, keeping earliest); (iii) duplicate IP address (2, keeping earliest); (iv) rating-phase duration under 2 minutes (12); (v) perfect straight-lining of preference ratings (2). After filtering, $N = 83$ raters contributed 4,150 ratings. Results were robust to filter choice: preference scores ranged from 0.58–0.63 on unfiltered data and from 0.62–0.68 on a stricter $>$ 3-minute filter.

Analysis and Results. We re-fitted the same Bradley–Terry model used in Table 2 (the paper's primary analysis), separately for each of the five dimensions using preference data from Clickworkers. Table 11 reports the results in the same format. Preference scores and odds ratios are within 0.05 and 0.35 of the original respectively for all five dimensions, with all comparisons highly significant. As a robustness check, we also re-fit logistic regression with cluster-robust standard errors clustering by rater (matching the paper's existing robustness check); estimates were substantively identical and are omitted for brevity.

Table 11: Bradley–Terry replication on $N = 83$ independent Clickworkers. Format matches Table 2. All p-values survive Holm correction at $\alpha = .05$.

Metric	n_{decisive}	β	SE	OR	PS	p
Clarity	748	0.491	0.075	1.63	.620	$< .001$
Contextual Fit	716	0.729	0.080	2.07	.675	$< .001$
Tone	686	0.481	0.079	1.62	.618	$< .001$
Enforceability	689	0.470	0.078	1.60	.615	$< .001$
Perceived Legitimacy	710	0.616	0.079	1.85	.649	$< .001$

C.4 Methods: Topic Coverage

Based on prior literature on online community governance, inspection of the mock conversation and analyzing the clusters in study 1, we defined 13 topic categories relevant to WhatsApp group moderation: *respect/civility*, *politics*, *spam/forwarding*, *privacy*, *conflict resolution*, *language/profanity*, *media sharing*, *relevance/on-topic*, *misinformation*, *new member protocols*, *private messaging*, *tone/-positivity*, and *sensitive topics*.

For each topic, we compiled a list of 3-7 keywords and keyword stems based on the most discriminative words for the topic and the cluster[4]. For example, the *politics* topic included the keywords: "politic," "political," "election," "vote," "party," and "government." The *privacy* topic included: "privacy," "private," "personal," "confidential," and "share outside." A rule set was coded as covering a topic if any of that topic's keywords appeared in the text (case-insensitive substring matching).

Table 12: Topic Coverage by Condition (Keyword-Based Detection). Here, pp = percentage points.

Topic	Manual (%)	AI-Assisted (%)	Difference (pp)
Spam / Forwarding	85.0	100.0	+15.0
Politics	80.0	100.0	+20.0
Language / Profanity	80.0	94.1	+14.1
Respectful Communication	50.0	94.1	+44.1
Privacy	25.0	91.2	**+66.2**
New Member Protocols	40.0	88.2	**+48.2**
Media Sharing	15.0	70.6	**+55.6**
Conflict Resolution	45.0	61.8	+16.8
Relevance / On-Topic	20.0	55.9	+35.9
Tone / Positivity	10.0	50.0	+40.0
Private Messages	30.0	47.1	+17.1
Sensitive Topics	15.0	41.2	+26.2
Misinformation	20.0	20.6	+0.6

C.5 Linguistic Markers for Prescriptive/Restrictive Classification

Following the framework of Fiesler et al. (2018), we classified linguistic markers into two categories based on their framing function:

Restrictive Markers. Language that explicitly limits or forbids actions:

- `no [X]` – e.g., "No political discussions", "No spam"
- `do not` – e.g., "Do not share personal information"
- `don't` – e.g., "Don't post advertisements"
- `avoid` – e.g., "Avoid heated debates"
- `refrain` – e.g., "Refrain from using profanity"
- `prohibited` – e.g., "Personal attacks are prohibited"

[4]We used TD-IDF scores to retrieve these words with all the rules in a cluster concatenated to form a document.

- `forbidden` – e.g., “Hate speech is forbidden”
- `not allowed` – e.g., “Spam is not allowed”
- `limit` – e.g., “Limit forwarded messages”
- `reduce` – e.g., “Reduce the number of images”
- `minimize` – e.g., “Minimize off-topic posts”
- `restrict` – e.g., “Restrict commercial content”
- `cannot / can't` – e.g., “You cannot share external links”
- `should not / shouldn't` – e.g., “Members shouldn't argue”

Prescriptive Markers. Language expressing guidelines or positive directives:

- `be [adjective]` – e.g., “Be respectful”, “Be polite”, “Be kind”
- `maintain` – e.g., “Maintain a respectful tone”
- `keep it [adjective]` – e.g., “Keep it positive”, “Keep it friendly”
- `treat` – e.g., “Treat everyone with respect”
- `inform` – e.g., “Inform the group before adding members”
- `announce` – e.g., “Announce new members”
- `verify` – e.g., “Verify information before sharing”
- `respect` – e.g., “Respect everyone's opinions”
- `ensure` – e.g., “Ensure all posts are relevant”
- `communicate` – e.g., “Communicate with kindness”
- `welcome` – e.g., “Welcome new members”
- `introduce` – e.g., “Introduce yourself to the group”
- `follow` – e.g., “Follow community guidelines”
- `share [noun]` – e.g., “Share family news”, “Share positive updates”

Markers were matched using case-insensitive regular expressions. Each participant's rules were analyzed to compute the proportion of restrictive markers (restrictive count / total markers), with prescriptive proportion as its complement.